

Mohamed Abumandour

Professor of Anatomy and Embryology

Gametogenesis

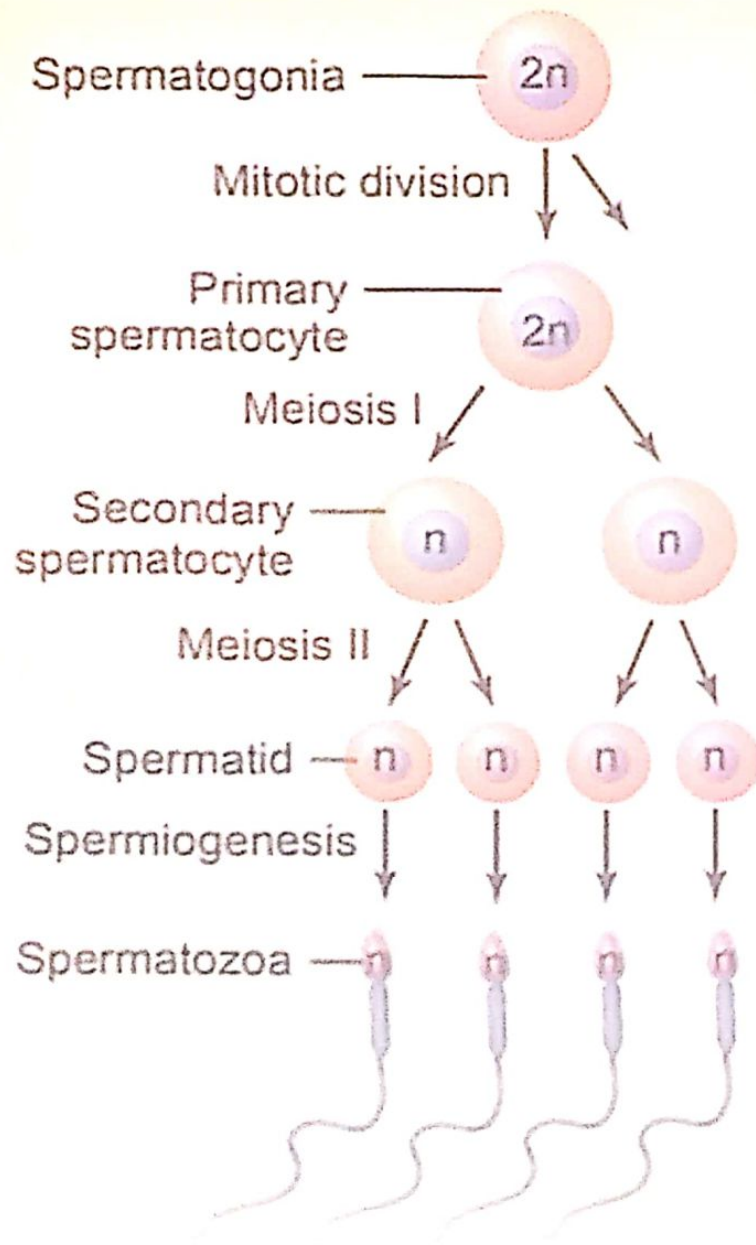
- Origin, Development, Maturation of Male & Female gametes (**Spermatogenesis and Oogenesis**)
- Process by which **diploid precursor cells ($2N$)** undergo **meiotic** division to become **haploid gametes (sex cells, N)**
- **In male:** process called **spermatogenesis** & produce spermatozoa (sperm)
- **In females:** process called oogenesis & produce ova (eggs)

Spermatogenesis

Multiplication
Phase

Growth
Phase

Maturation
Phase



Oogenesis

Foetal
life

Adult
reproductive
life

Second
polar
body

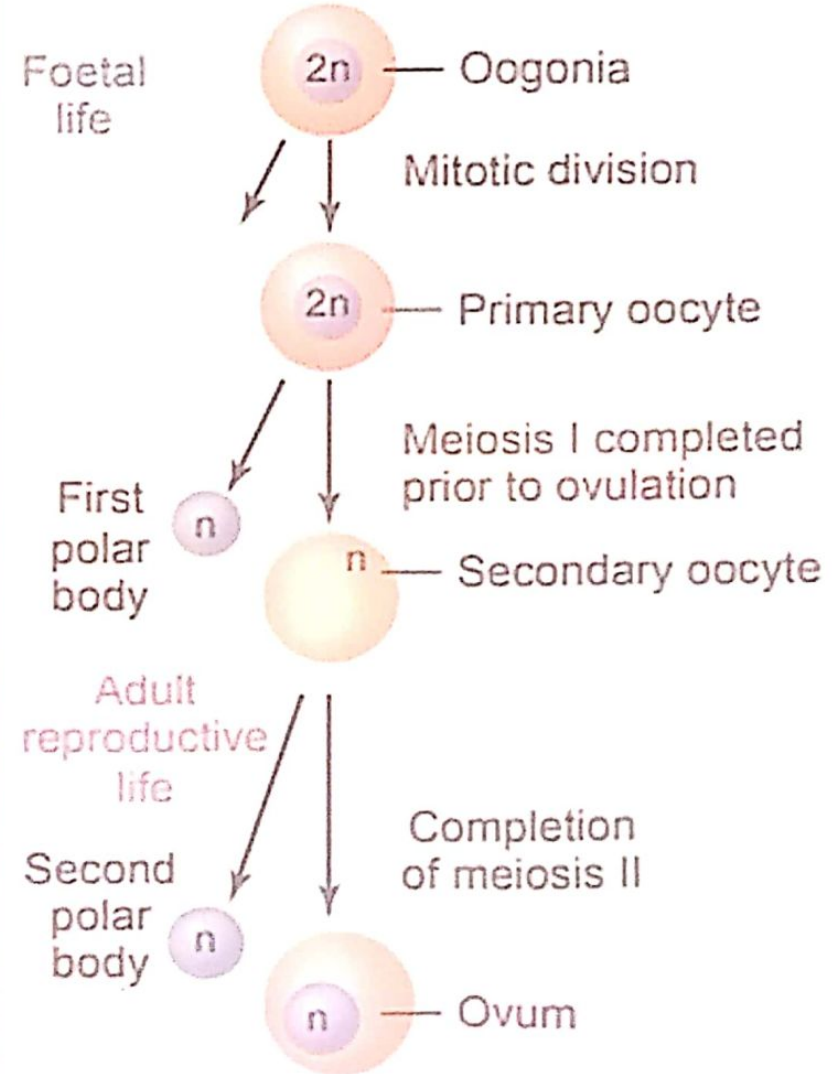


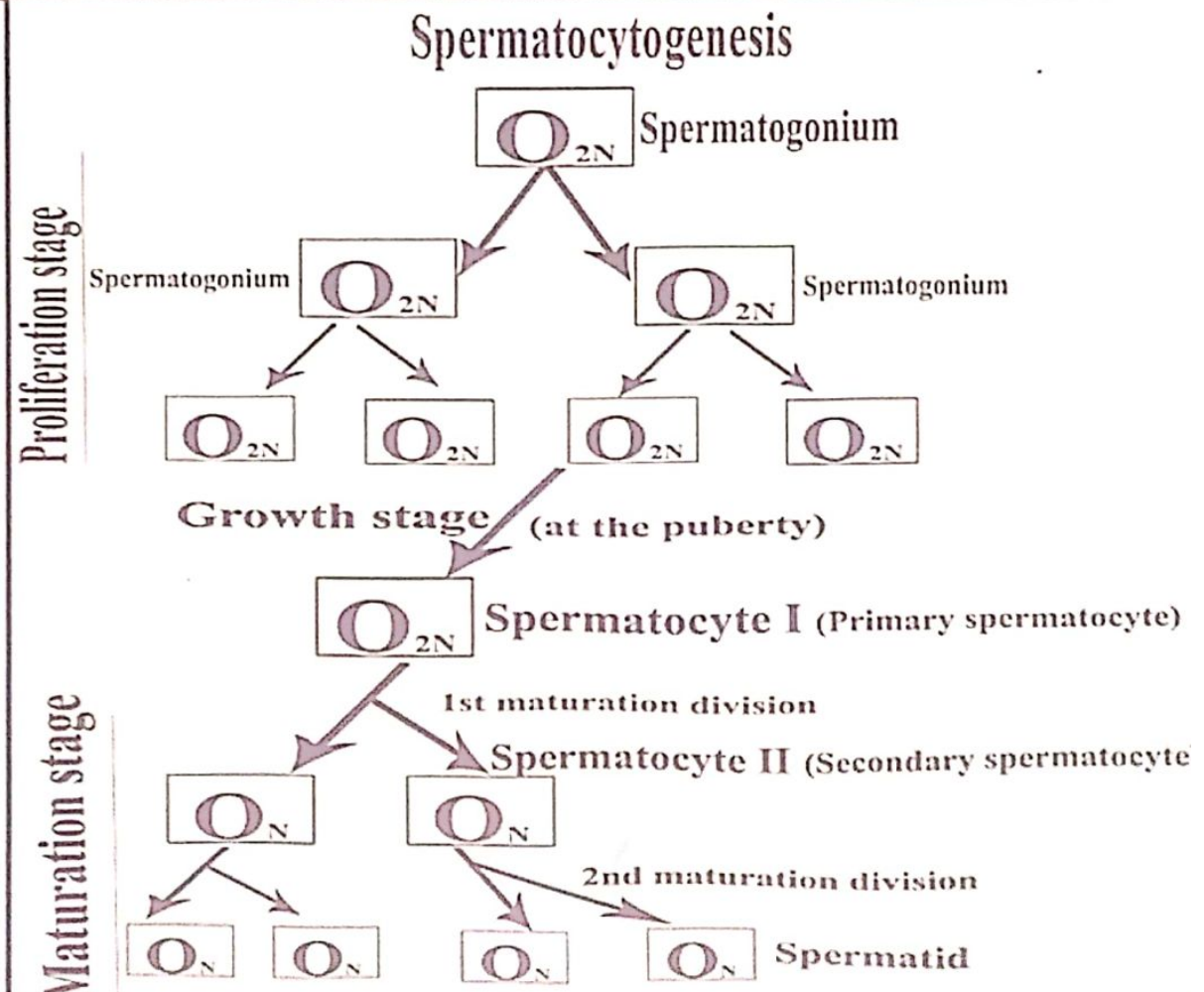
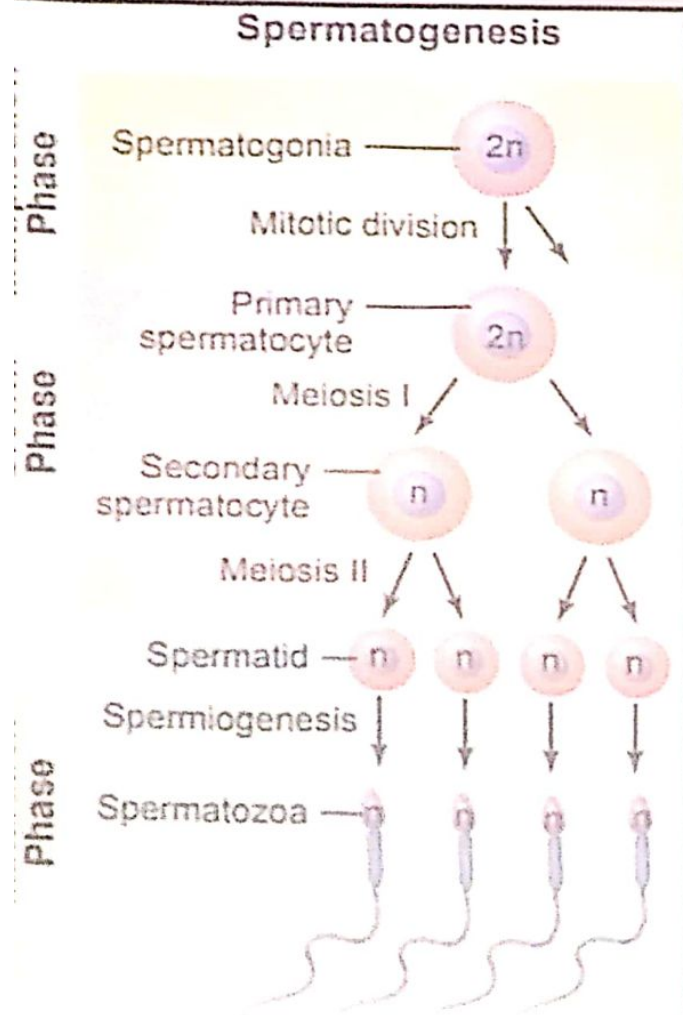
Fig. 2.5 Gametogenesis

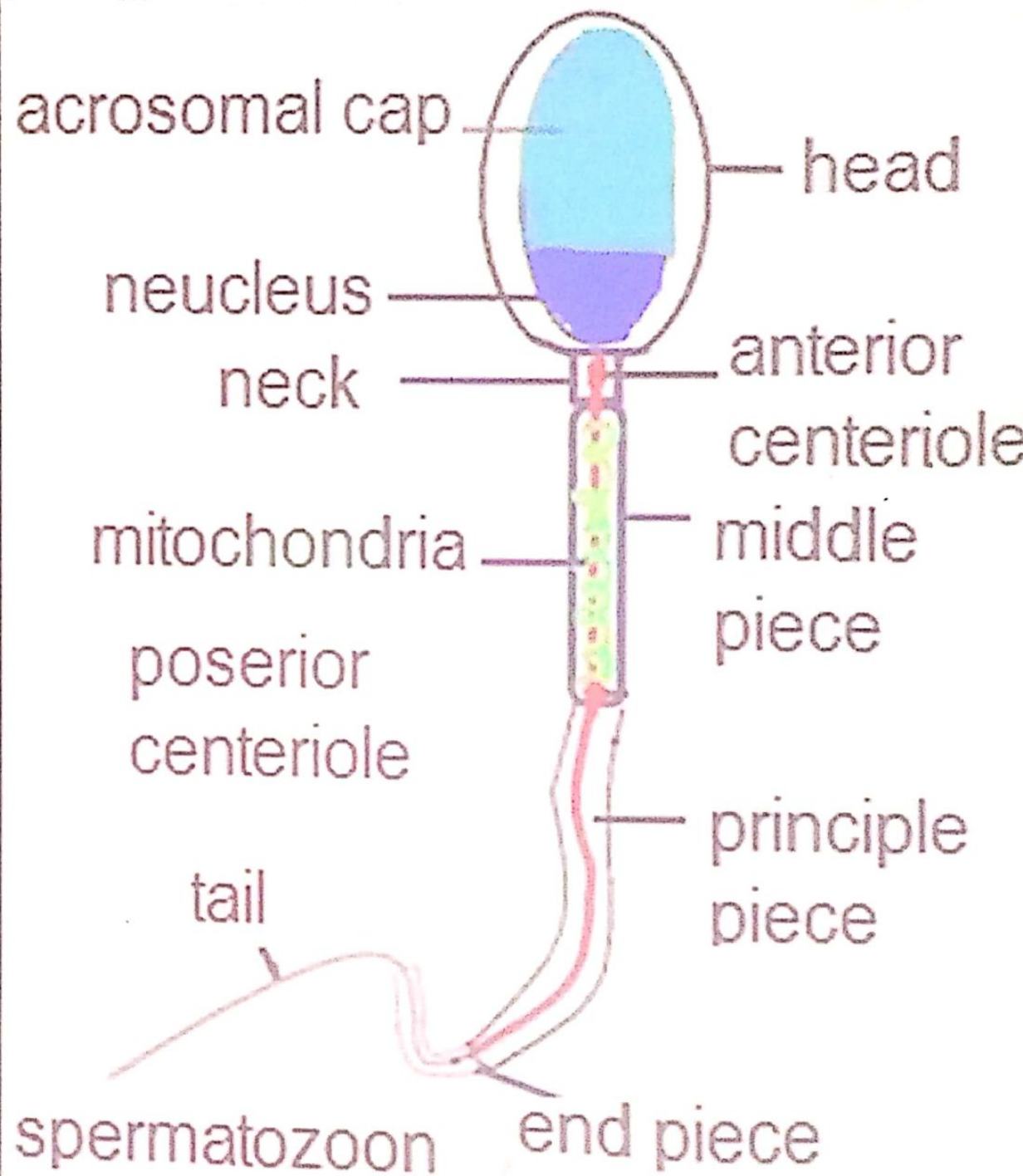
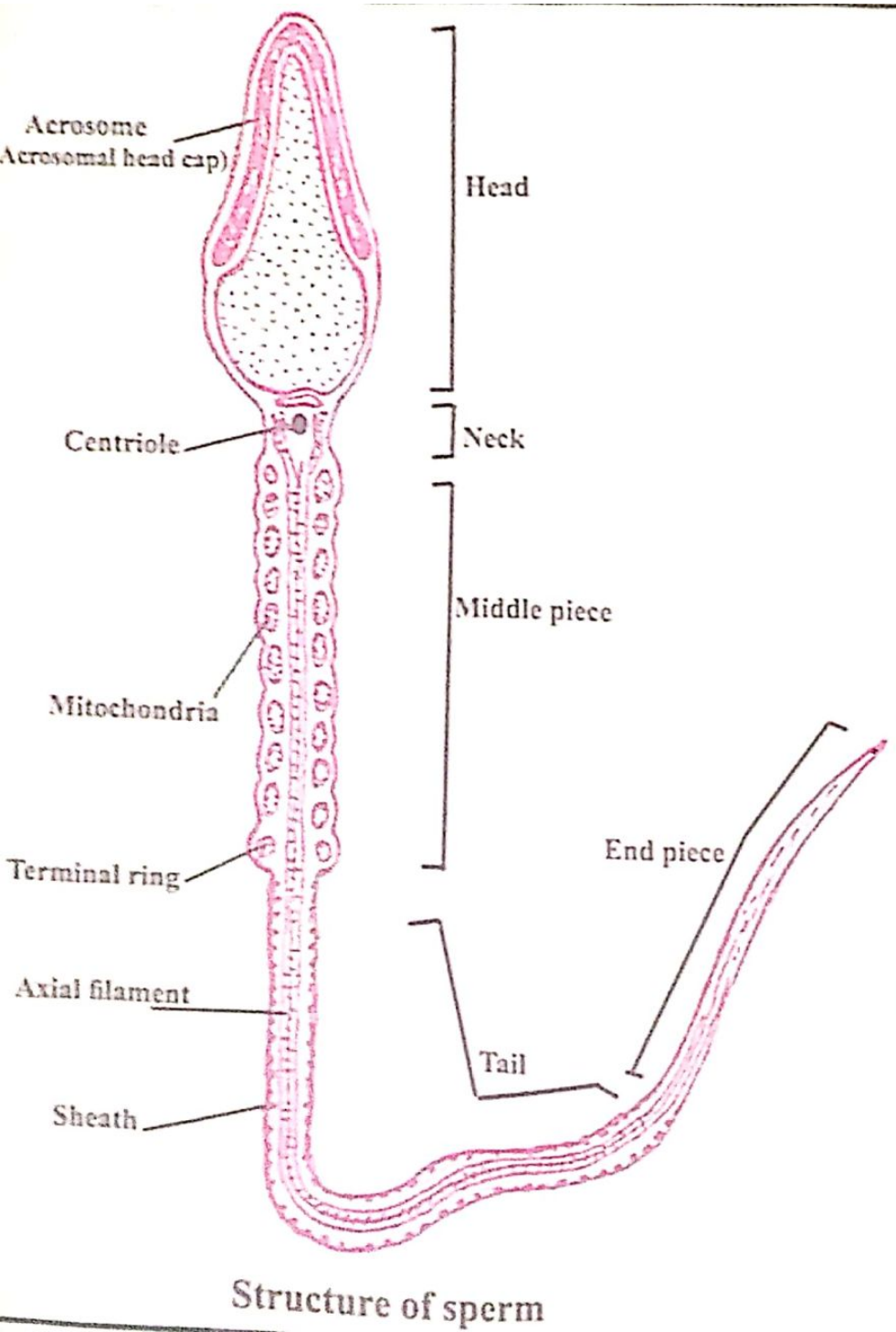
Spermatogenesis

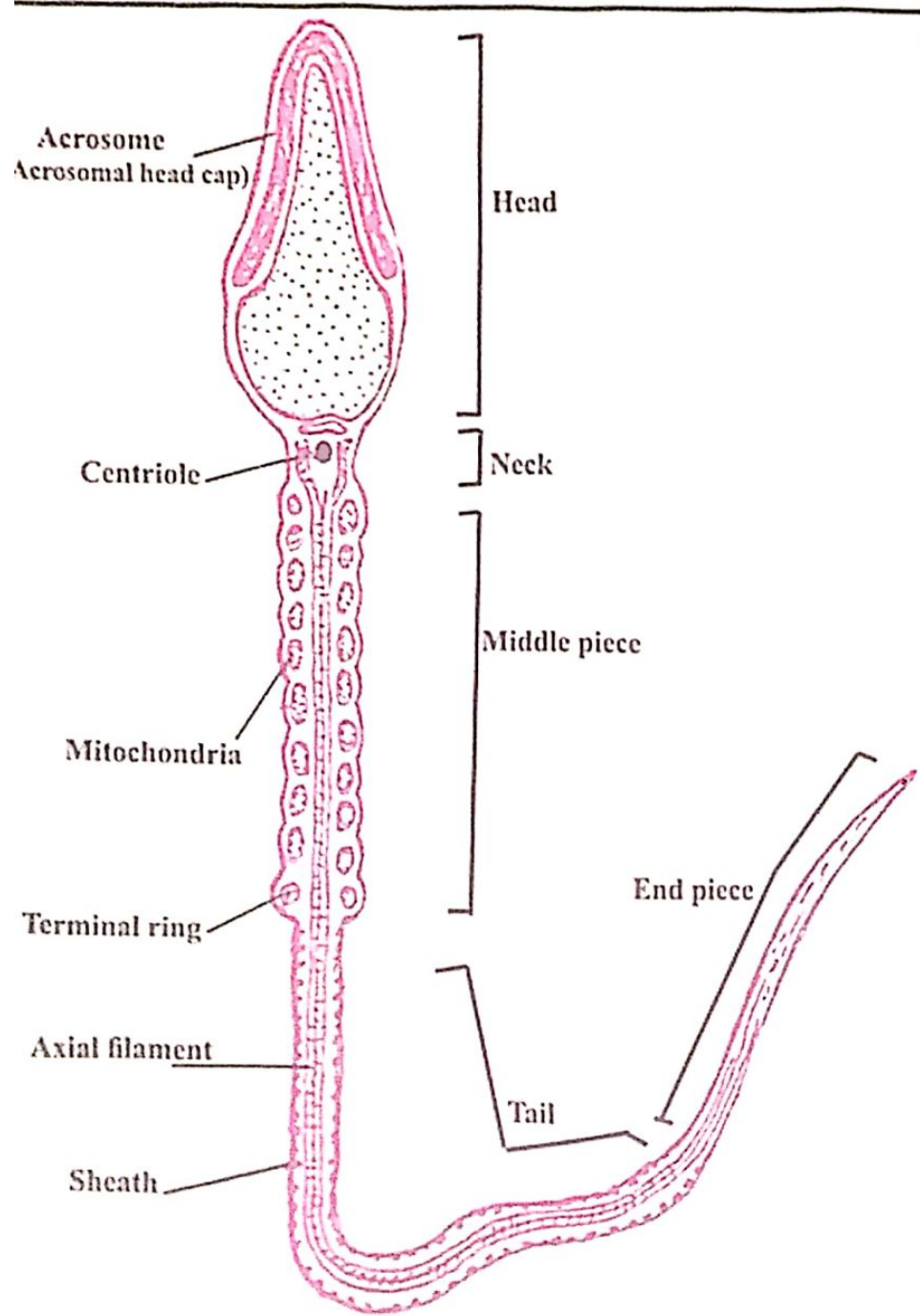
- Origin, development and Maturation of the sperm.

- Includes two processes

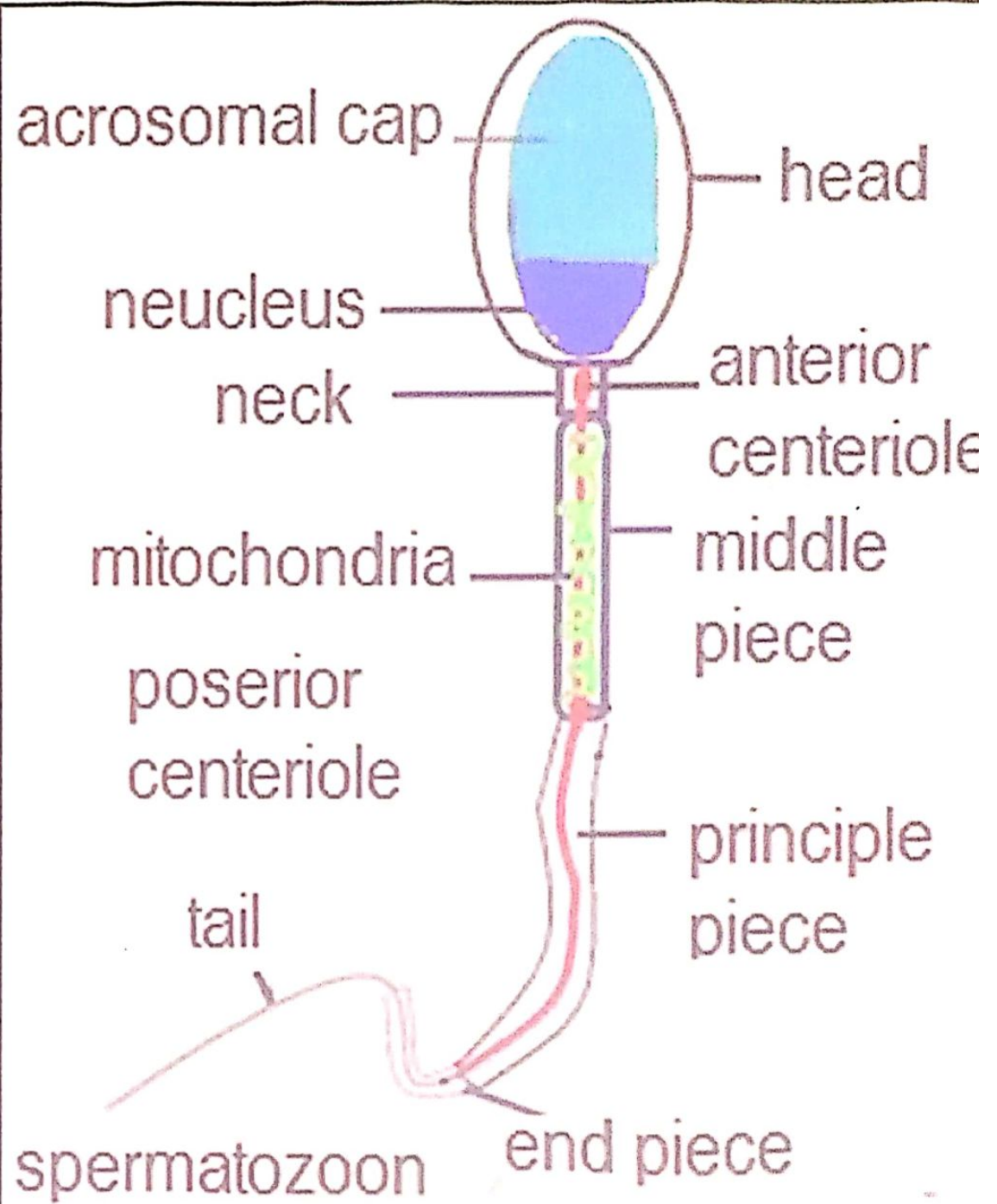
1. **Spermatocytogenesis**: Formation of spermatid from spermatogonia
2. **Spermatohistogenesis or Spermiogenesis**: Transformation of non motile spermatid into motile spermatozoon

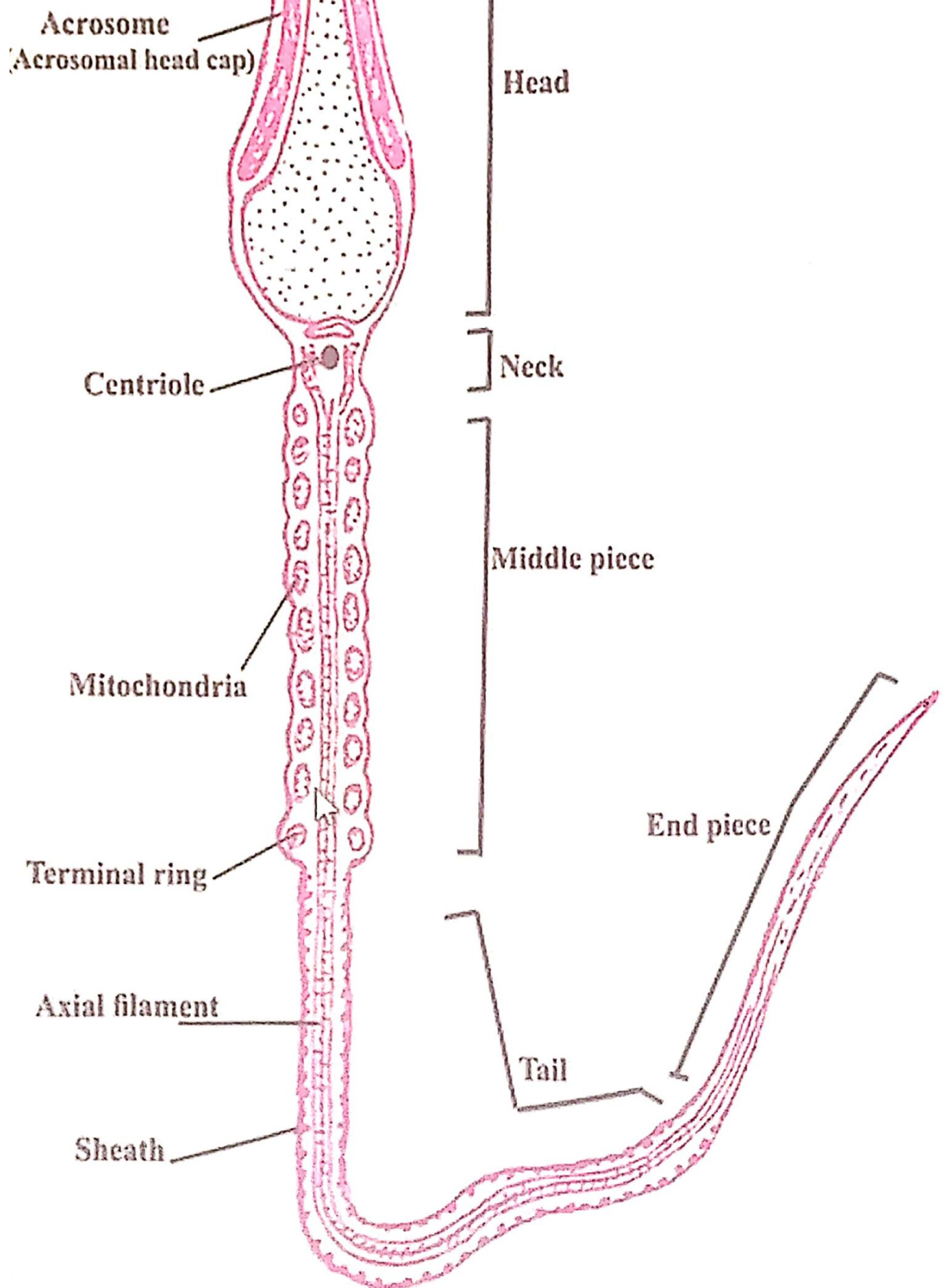




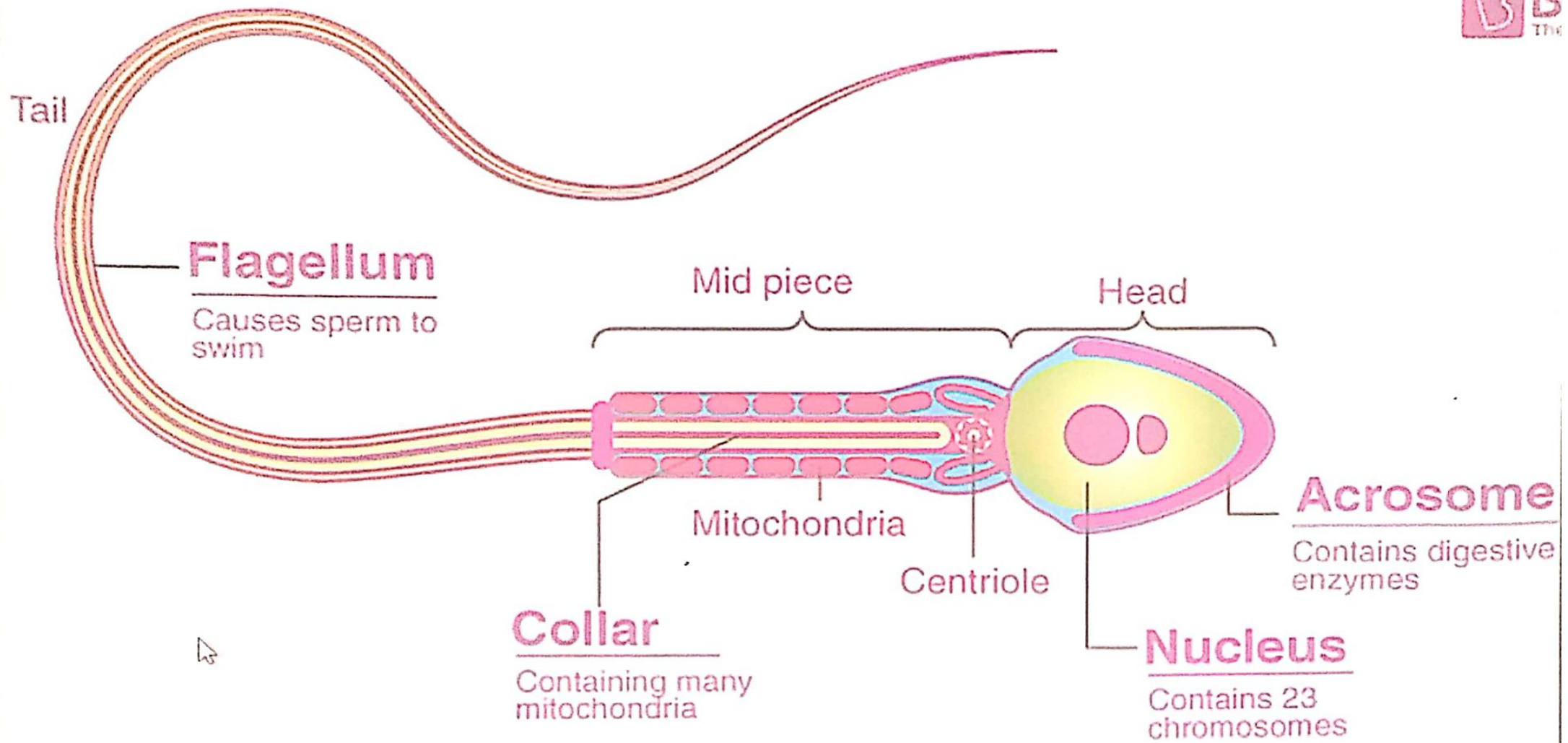


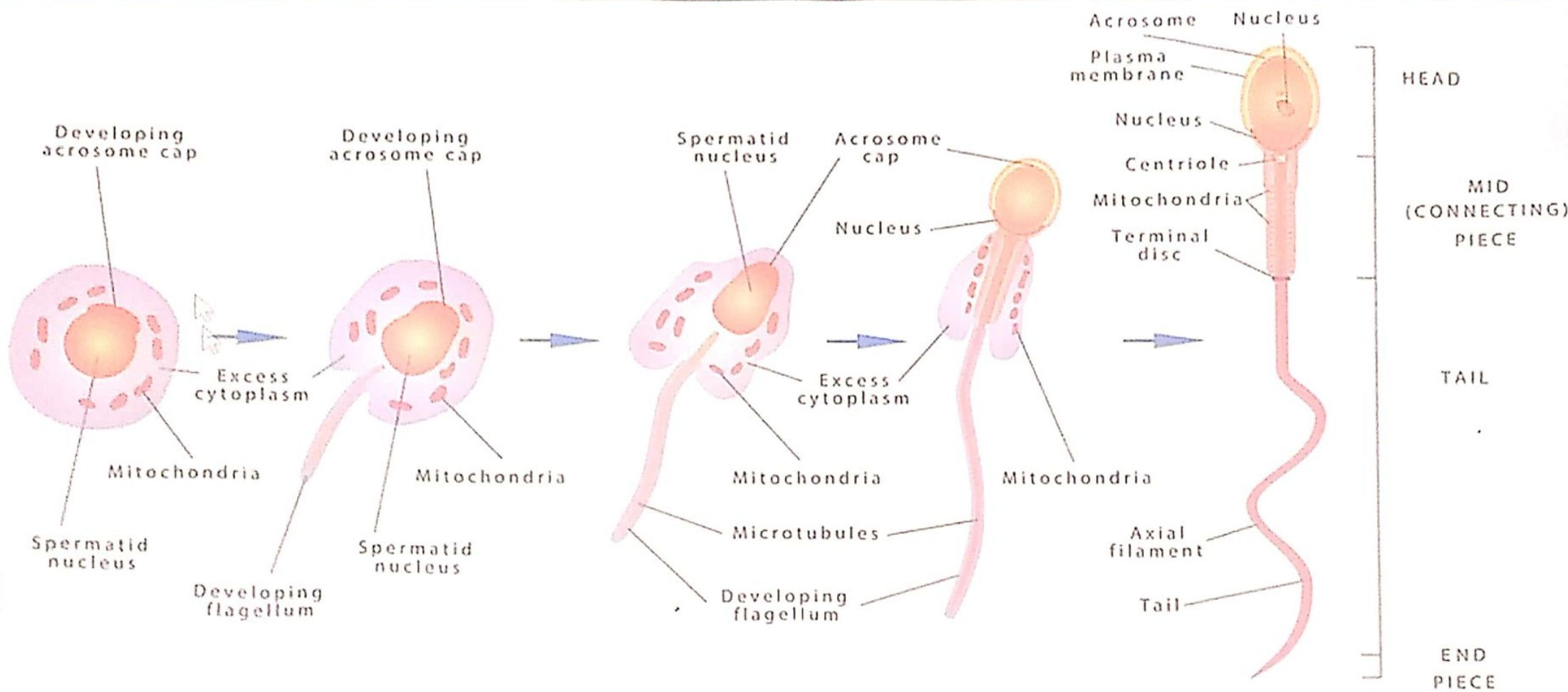
Structure of sperm



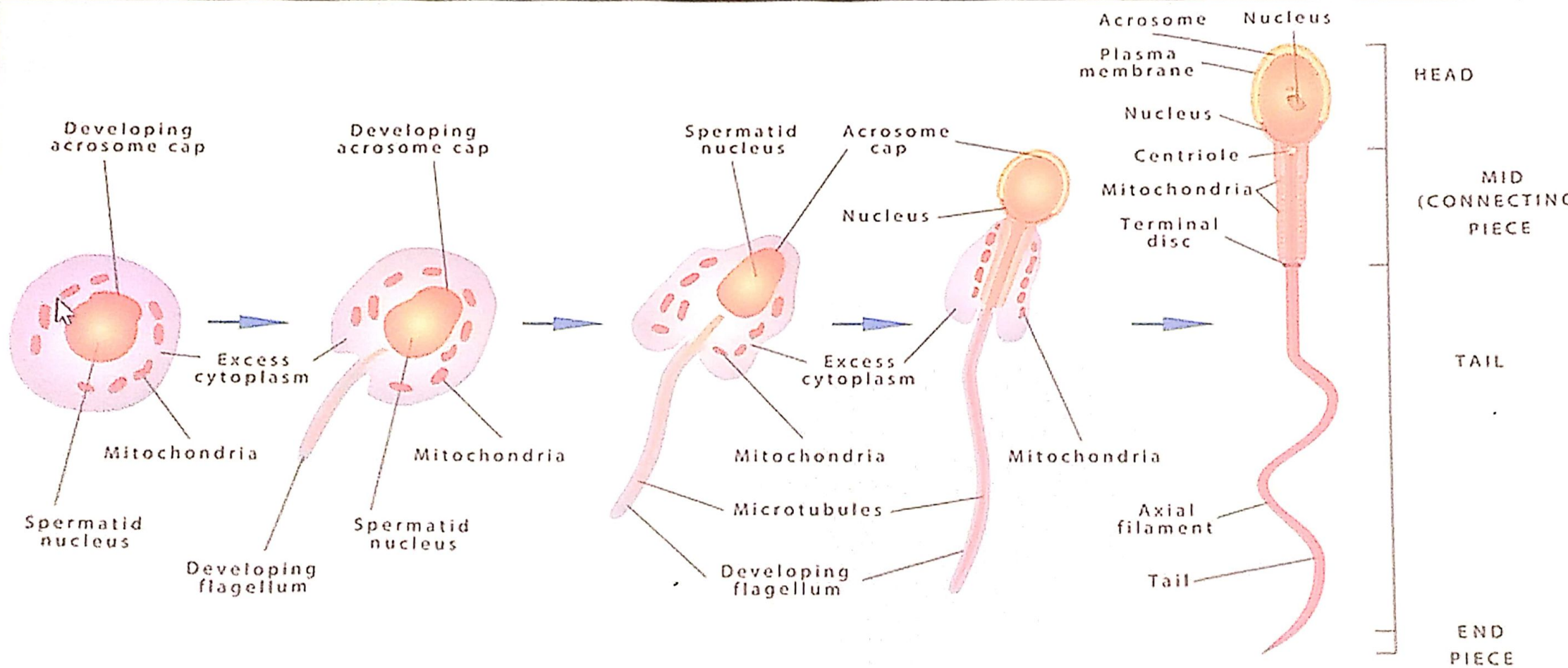


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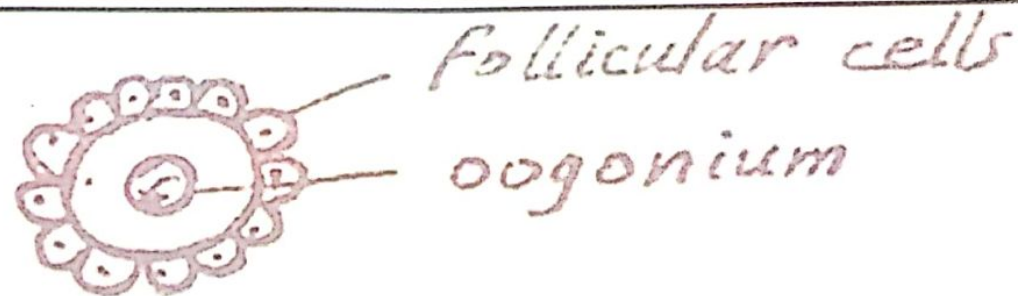
THE DEVELOPMENT OF SPERM



THE DEVELOPMENT OF SPERM

Oogenesis

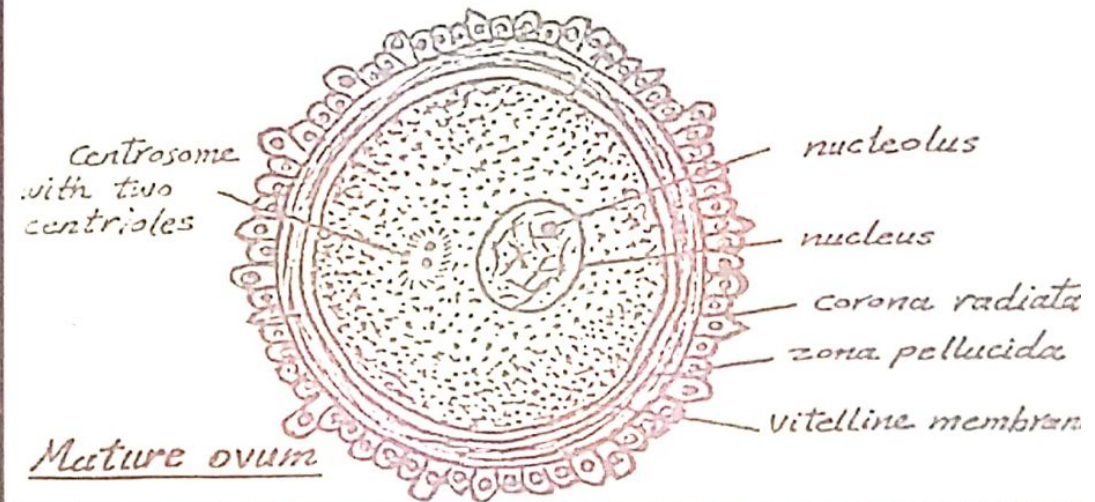
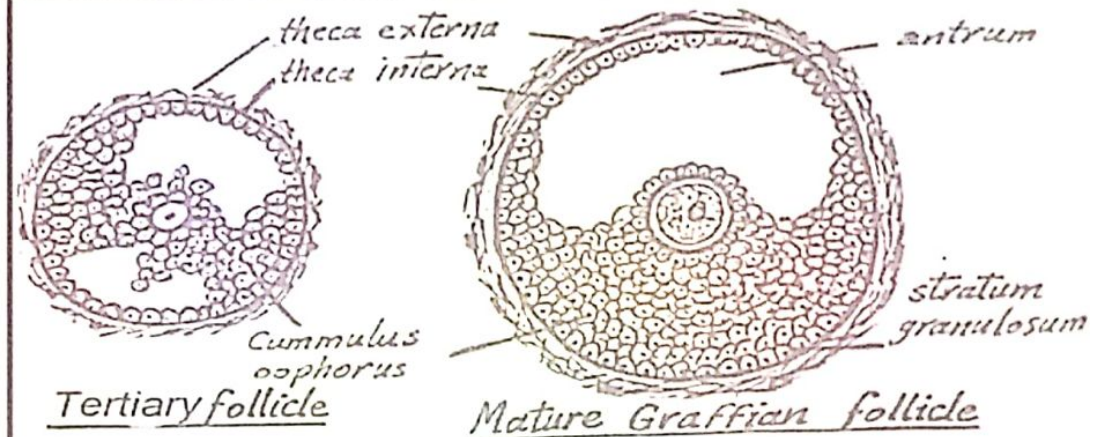
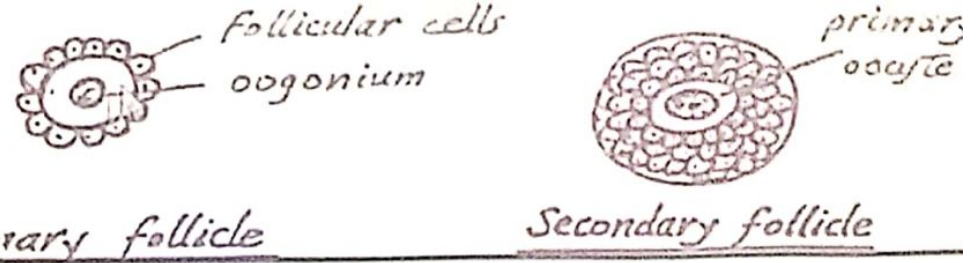
- It is the formation of the mature ova.
- **Oogonium:** migrated from endoderm of yolk sac through mesentery to localized in the **individual gonads**
- **Oogonium:** proliferated in ovary during intra-uterine life .
- Before birth:
 - ✓ All oogonium proliferated in ovary of fetus.
- After birth:
 - ✓ All oogonium transform to **primary follicle** (surrounded by one layer of follicular cells).

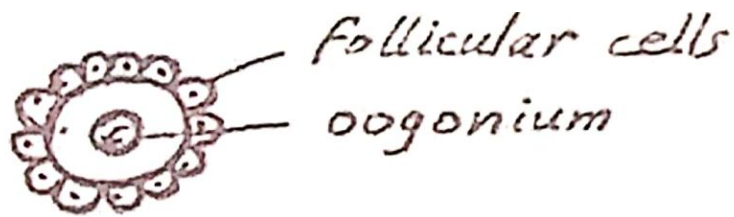


Primary follicle

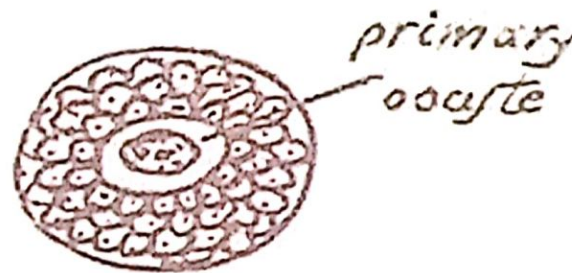
At puberty:

- ✓ Primary follicles enlarge to become secondary follicles (surrounded by more than one layer).
- ✓ Tertiary follicle (follicular cells forms small cavities between cells).
- ✓ Mature Graafian follicle (small cavities aggregated to form one large cavity filled by liquor fluid).

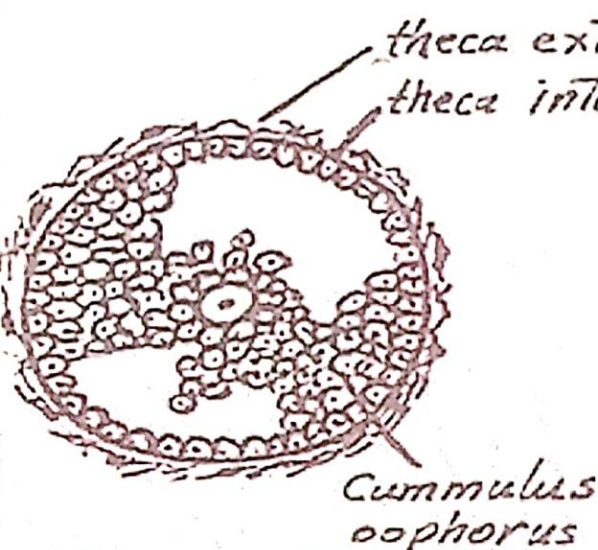




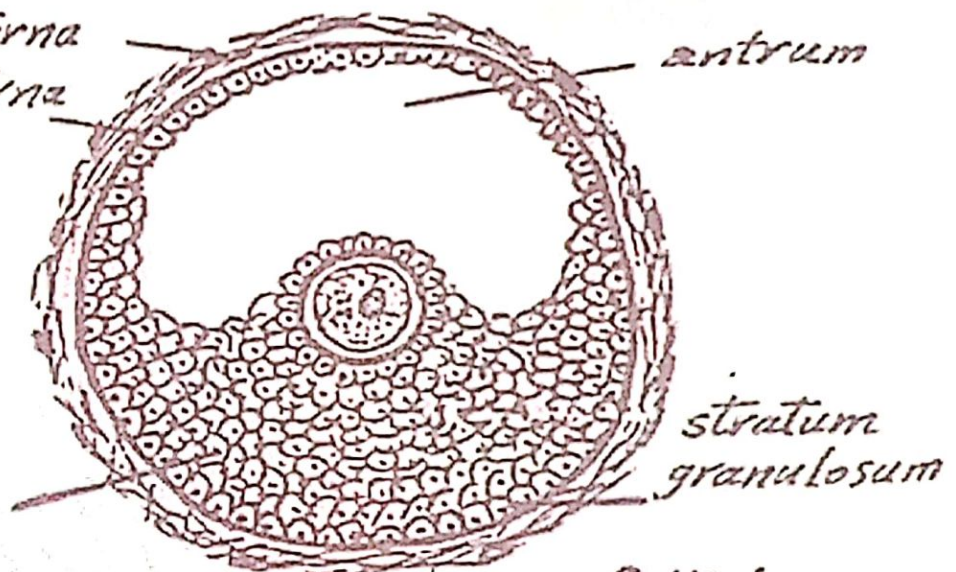
Primary follicle



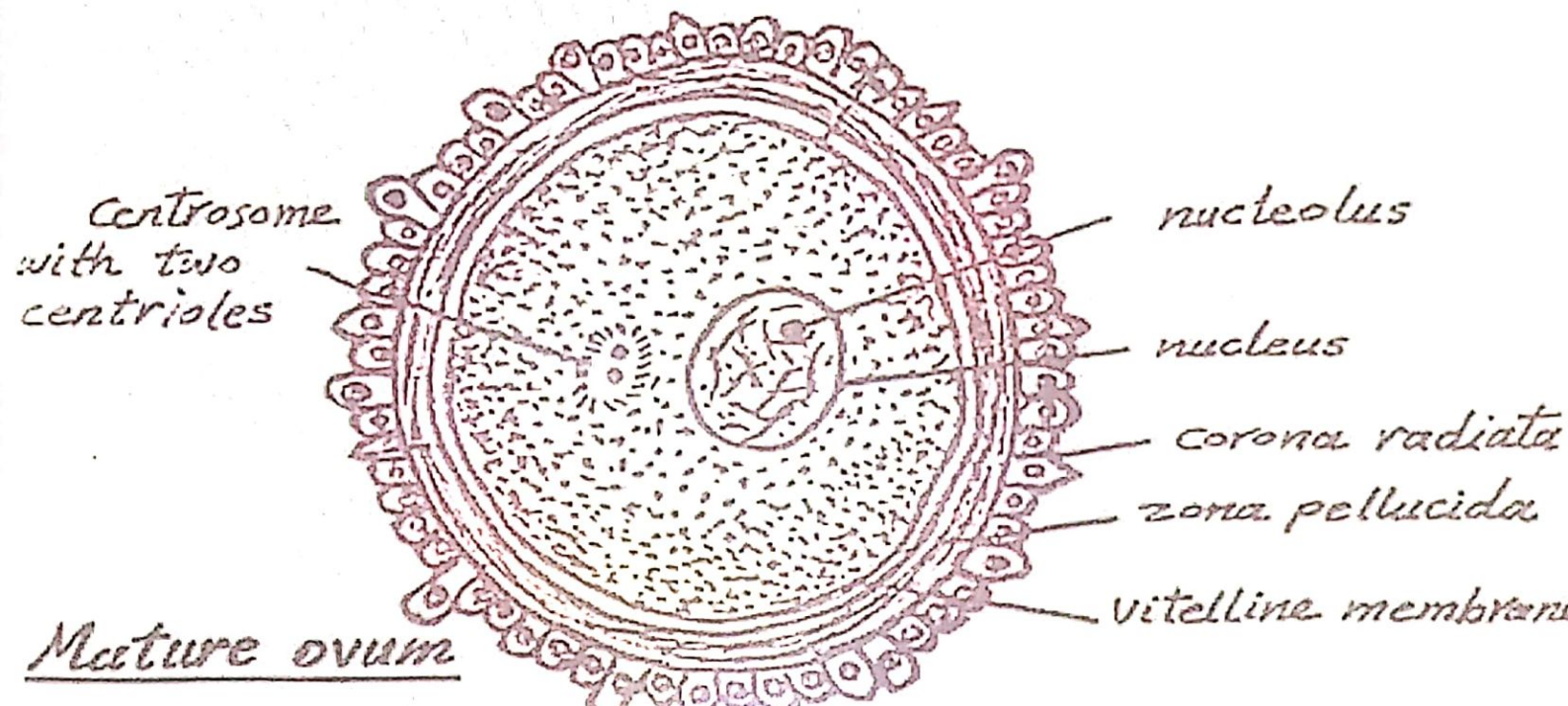
Secondary follicle



Tertiary follicle



Mature Graffian follicle



Mature ovum

Centrosome
with two
centrioles

nucleolus

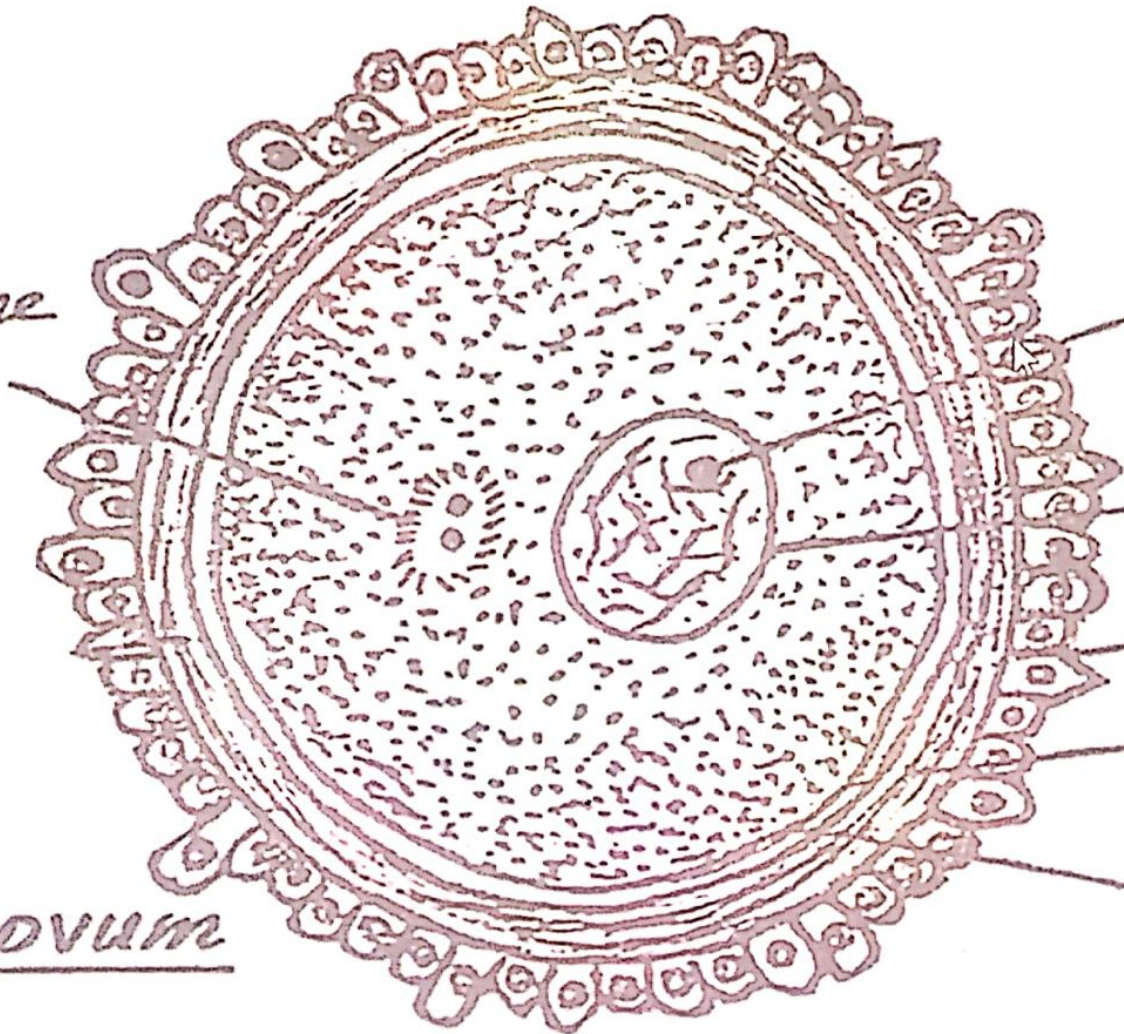
nucleus

corona radiata

zona pellucida

vitelline membrane

Mature ovum



Centrosome
with two
centrioles

nucleolus

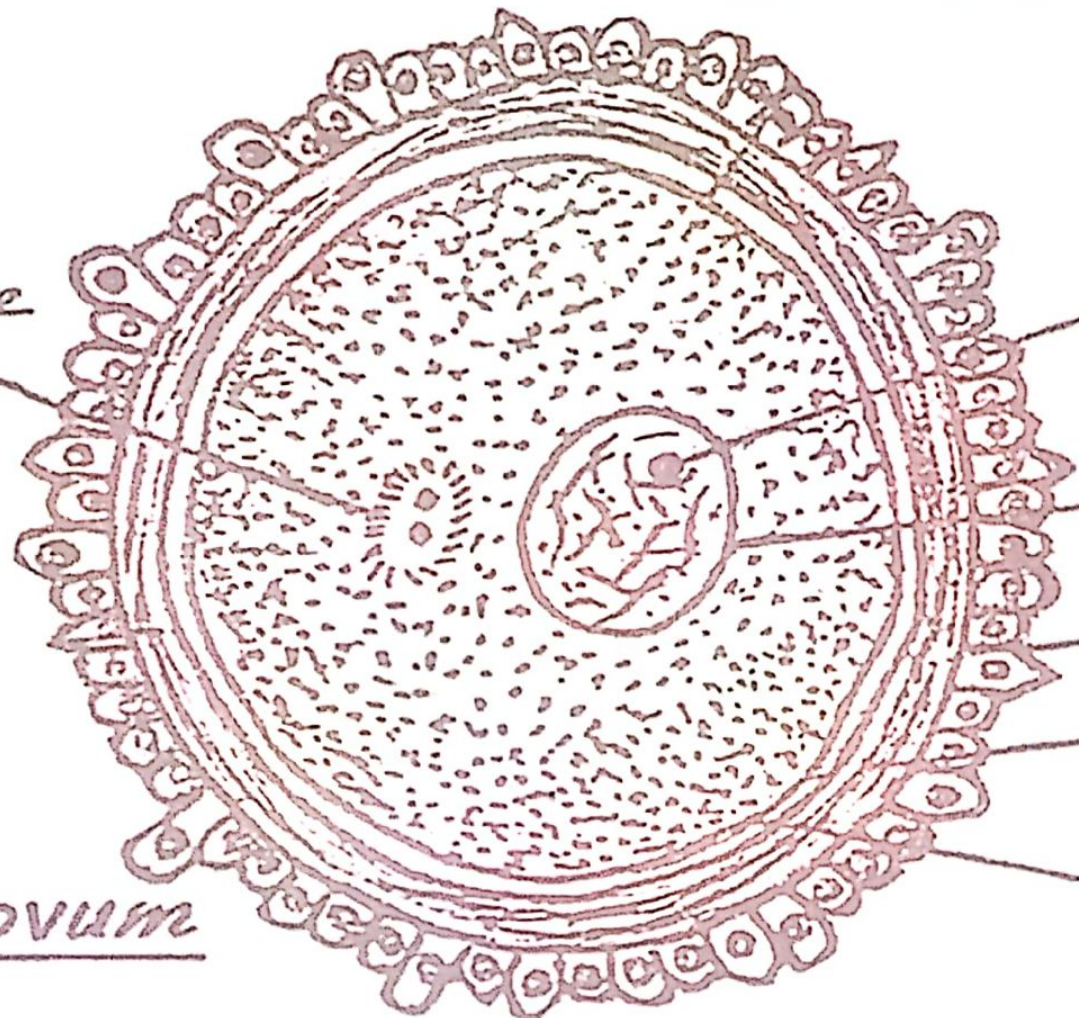
nucleus

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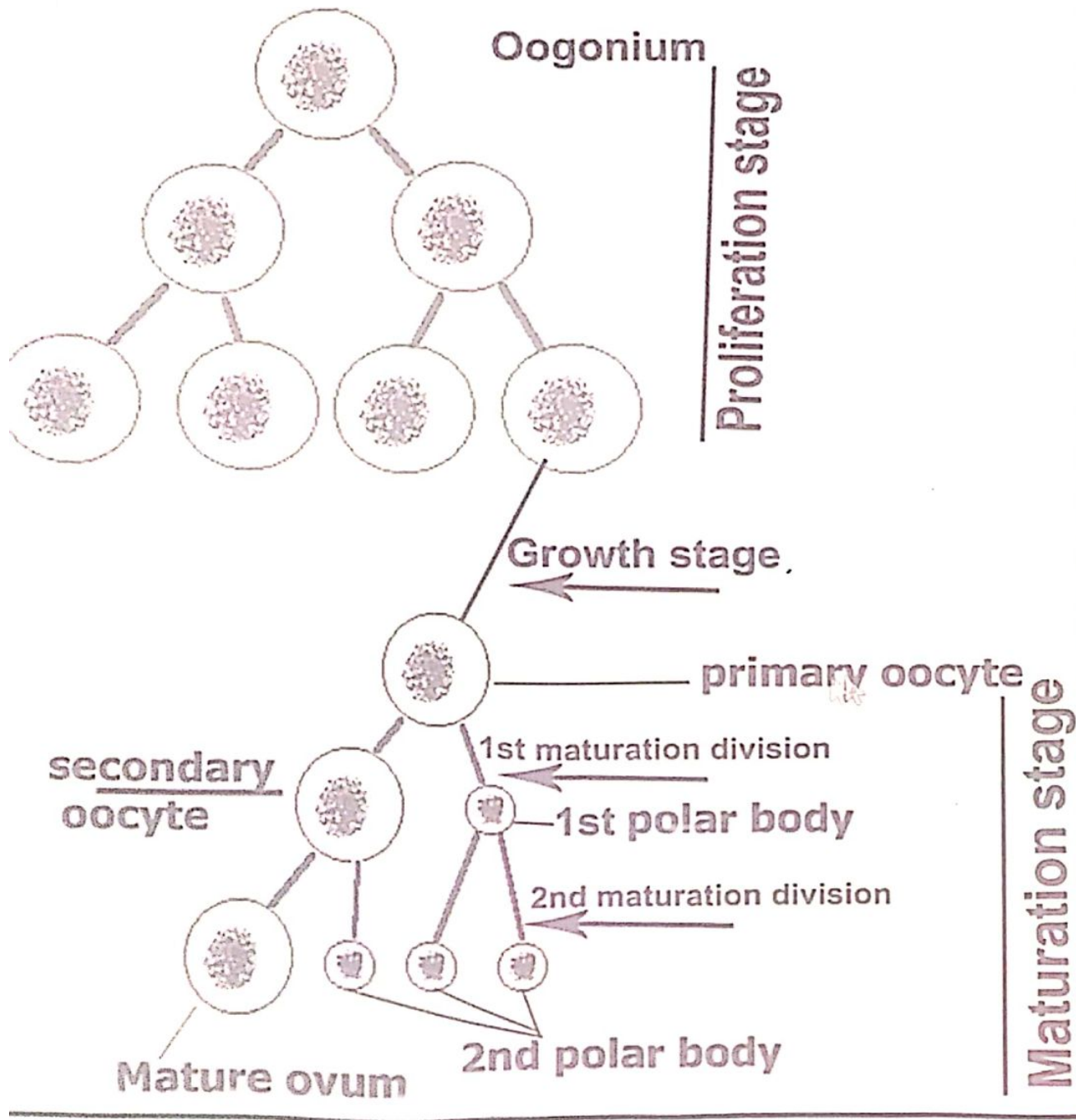
zona pellucida

vitelline membrane

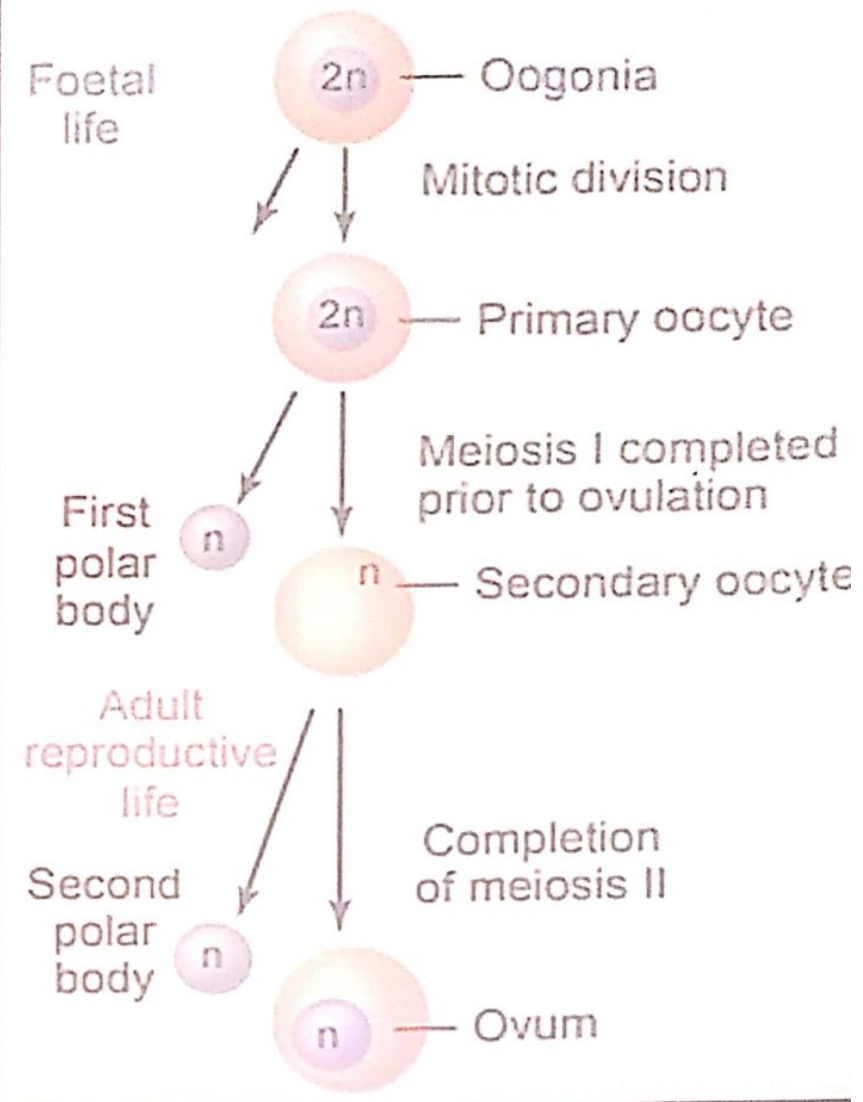
Mature ovum



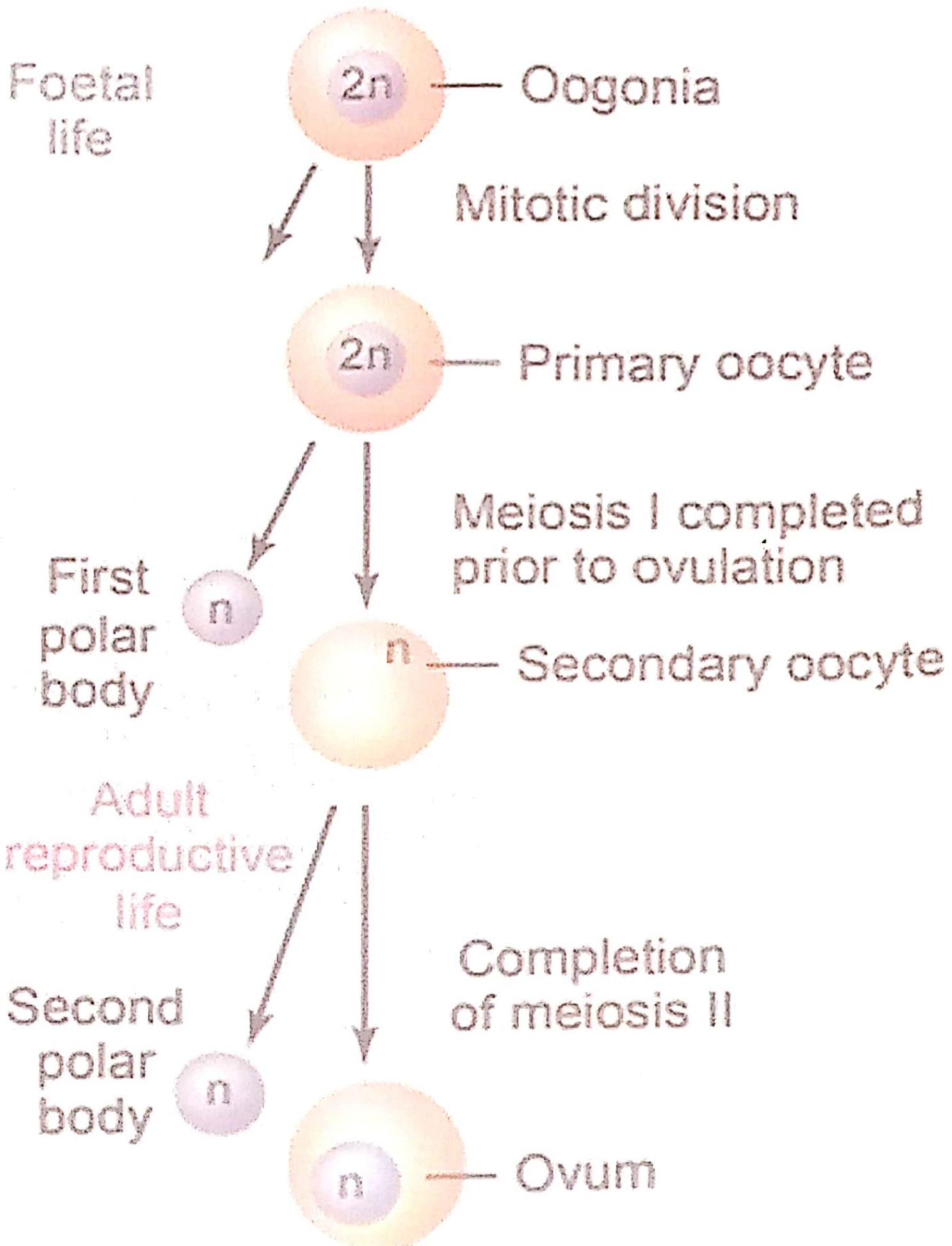
Oogenesis



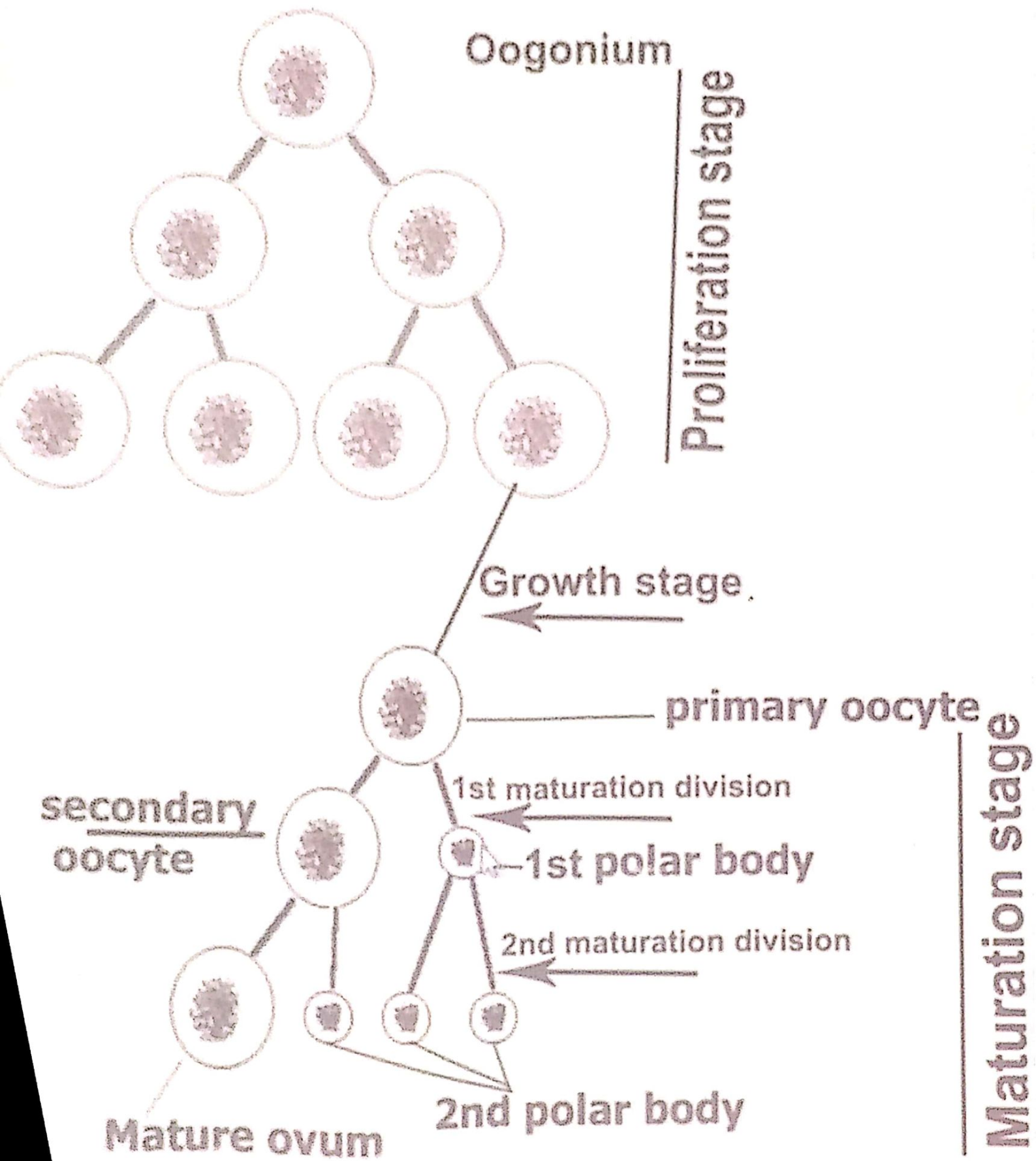
Oogenesis



Oogenesis

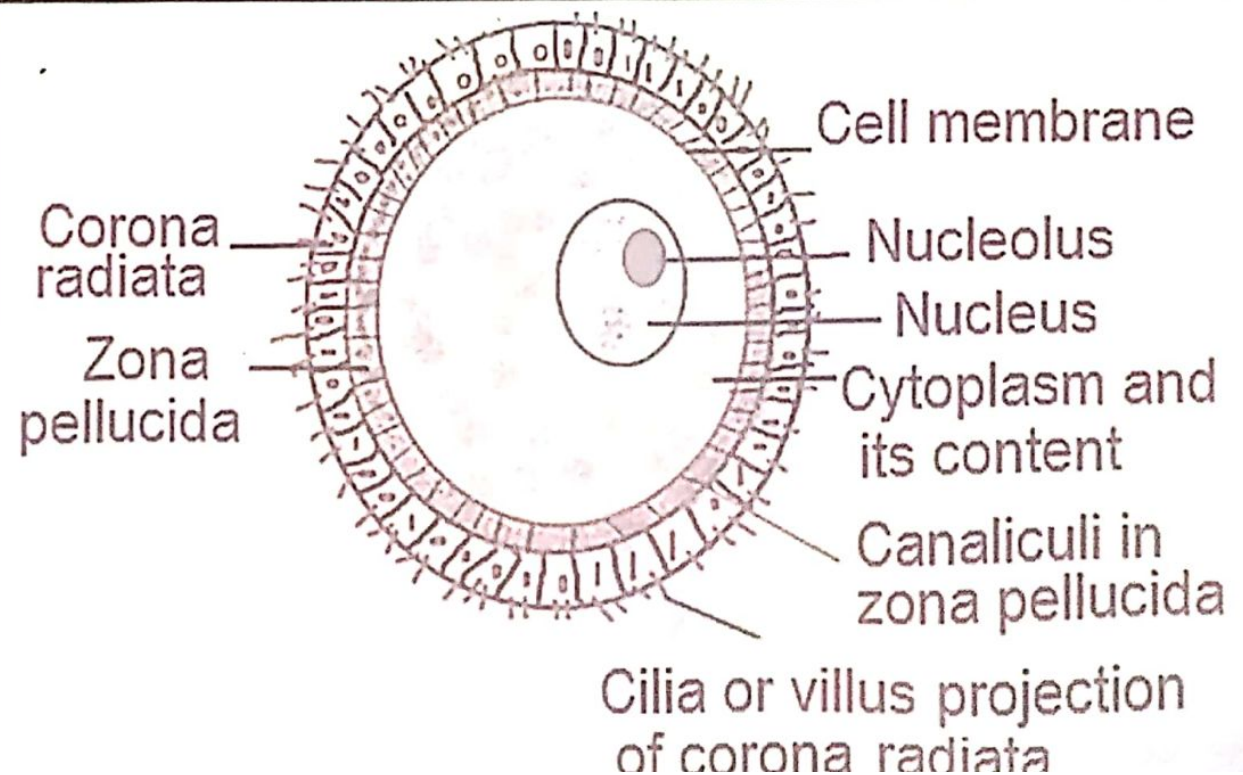
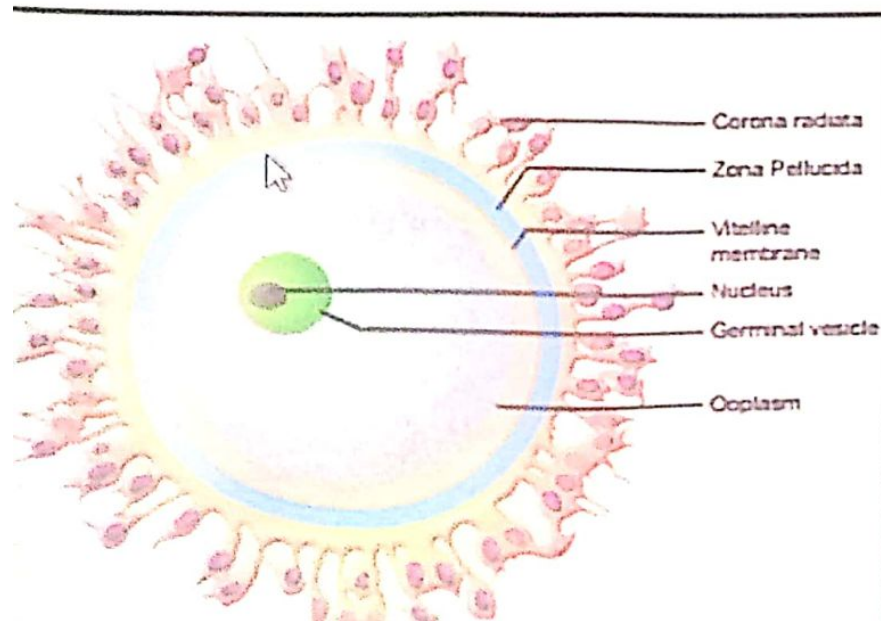


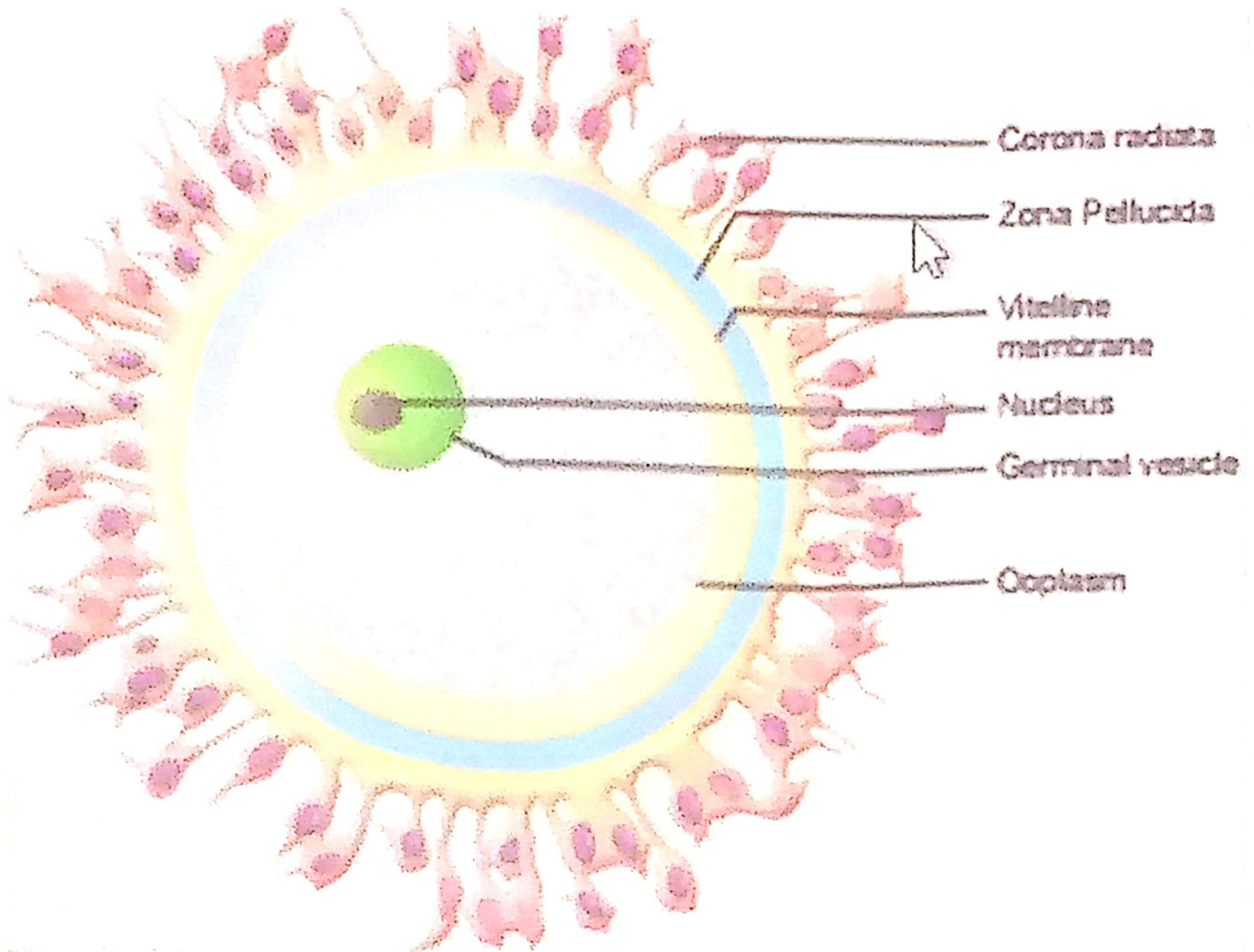
Oogenesis



Membrane of the ova

- **Primary membrane (inner)**: surrounded cytoplasm of ova called cell or cytoplasmic or vitelline membrane or oolemma.
- **Secondary membrane**: (zona pellucidae)
- **Tertiary membrane**:
 - Albumen layer around the rabbit egg .
 - Jelly-like material in frog ova.
 - Albumen layer and shell membrane in birds and reptile





Haploid Nucleus:

contains one copy of each chromosome.

Cytoplasm

Mitochondrion (pl.

mitochondria): produces ATP used for energy; contains mitochondrial DNA, which is passed to offspring.

Corticle Granule:

Releases enzymes during fertilization that harden the zona pellucida and digest binding proteins on sperm to prevent polyspermy after a sperm cell has entered the ovum.

First Polar Body

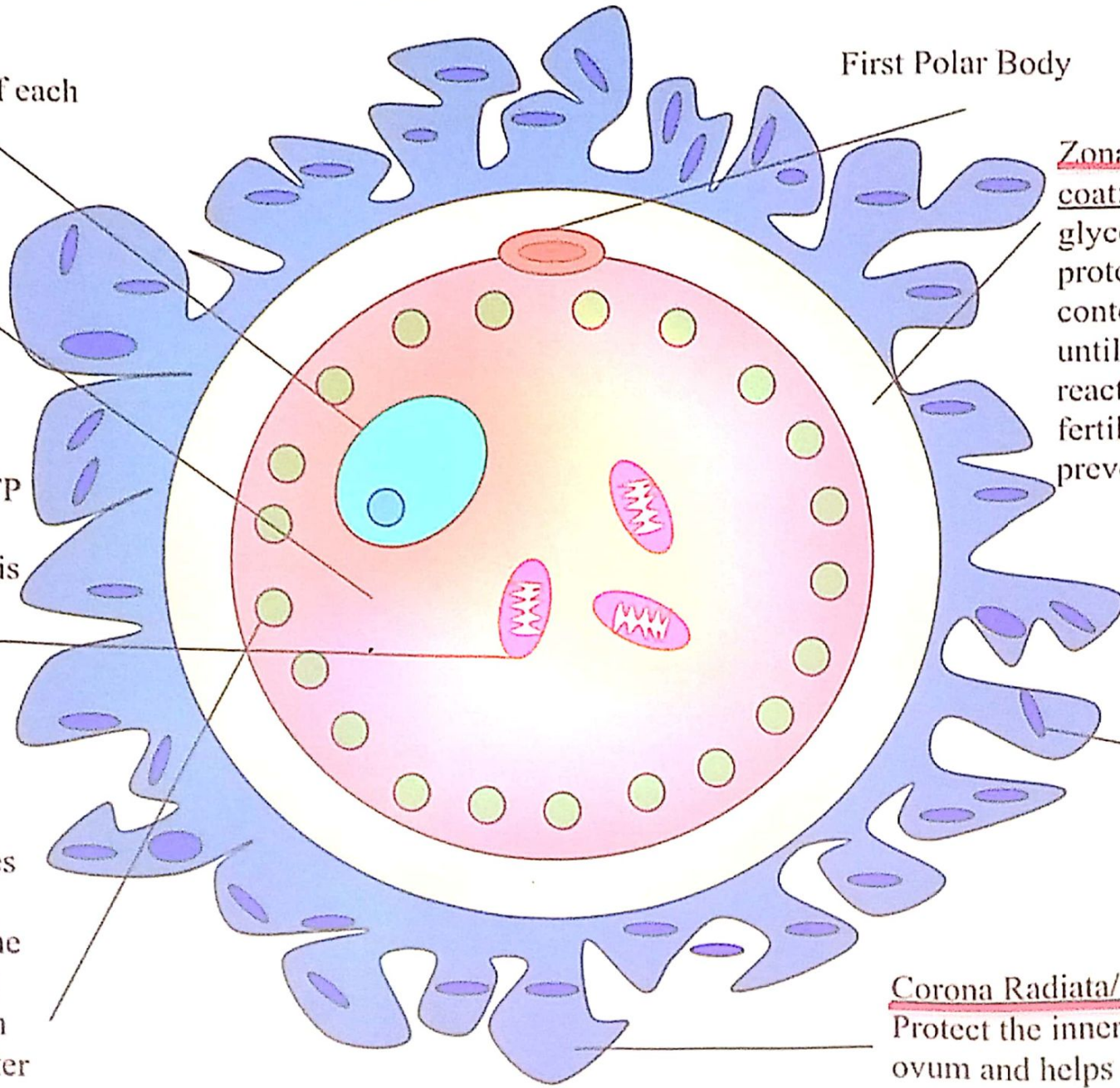
Zona Pellucida/jelly

coat: Contains glycoproteins that protect the inner contents of the ovum until the acrosome reaction of fertilization; helps prevent polyspermy.

Nucleus of Follicular Cell

Corona Radiata/follicular cells:

Protect the inner layers of the ovum and helps prevent polyspermy.

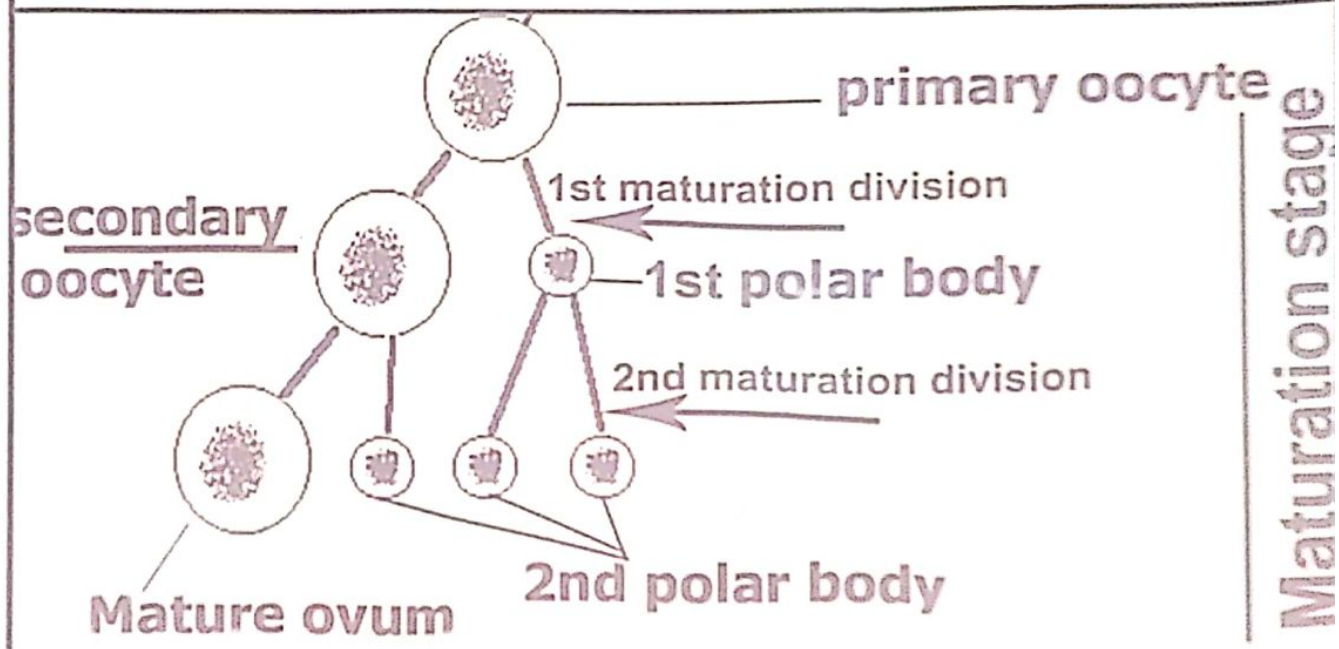
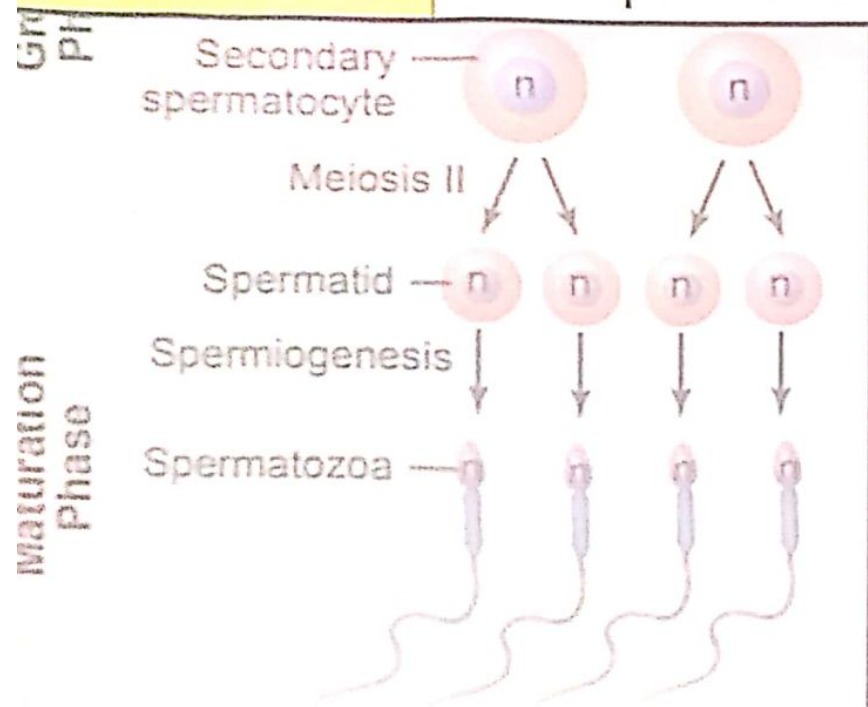


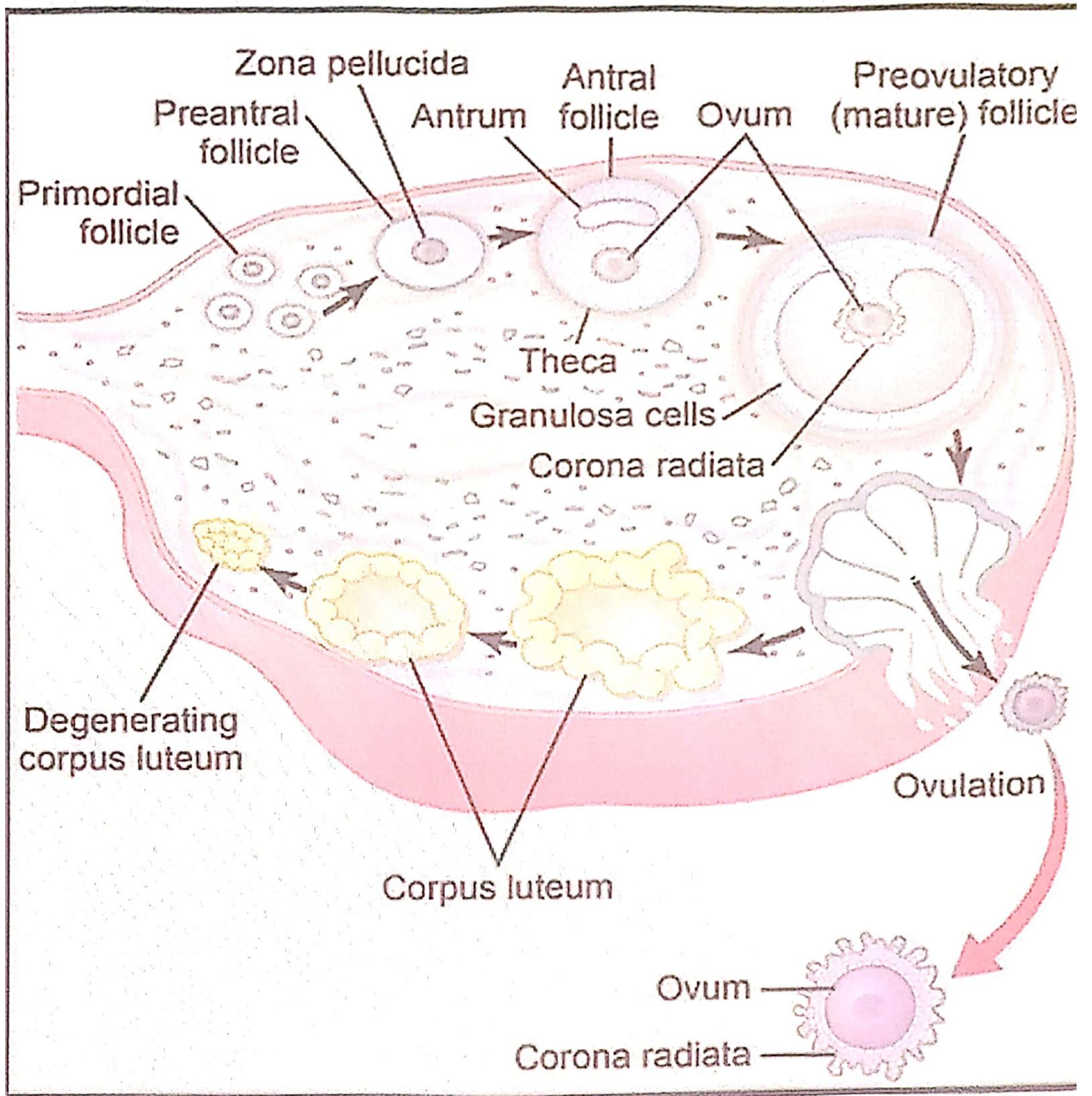
Egg (ovum) coverings

1) Primary	2) Secondary	3) Tertiary
Vitelline = plasma membrane	Zona pellucida (ZP)	a. jelly like material. b. Albumin c. albumin, shell membranes and shell
Formed by oocyte	Formed by follicular cells	formed by female genital tract when egg pass through it
Has role in fertilization	Has role in fertilization	• No role in fertilization • But just a protection. • Adhere eggs when reach water (a).
Present in all animals	In mammals only	a. in frog b. in rabbit c. in bird

Comparative between Spermatogenesis and Oogenesis

	Spermatogenesis	Oogenesis
Location	• Testis	• Ovary
Proliferation	• Long period • to the end of life	• Short period • Intra-uterine only
Growth	• Short	• Long
Stages	• Spermatocytogenesis • Spermiogenesis	• Oogenesis
End result	• 4 Sperm of equal size	• one ova + 3 polar bodies



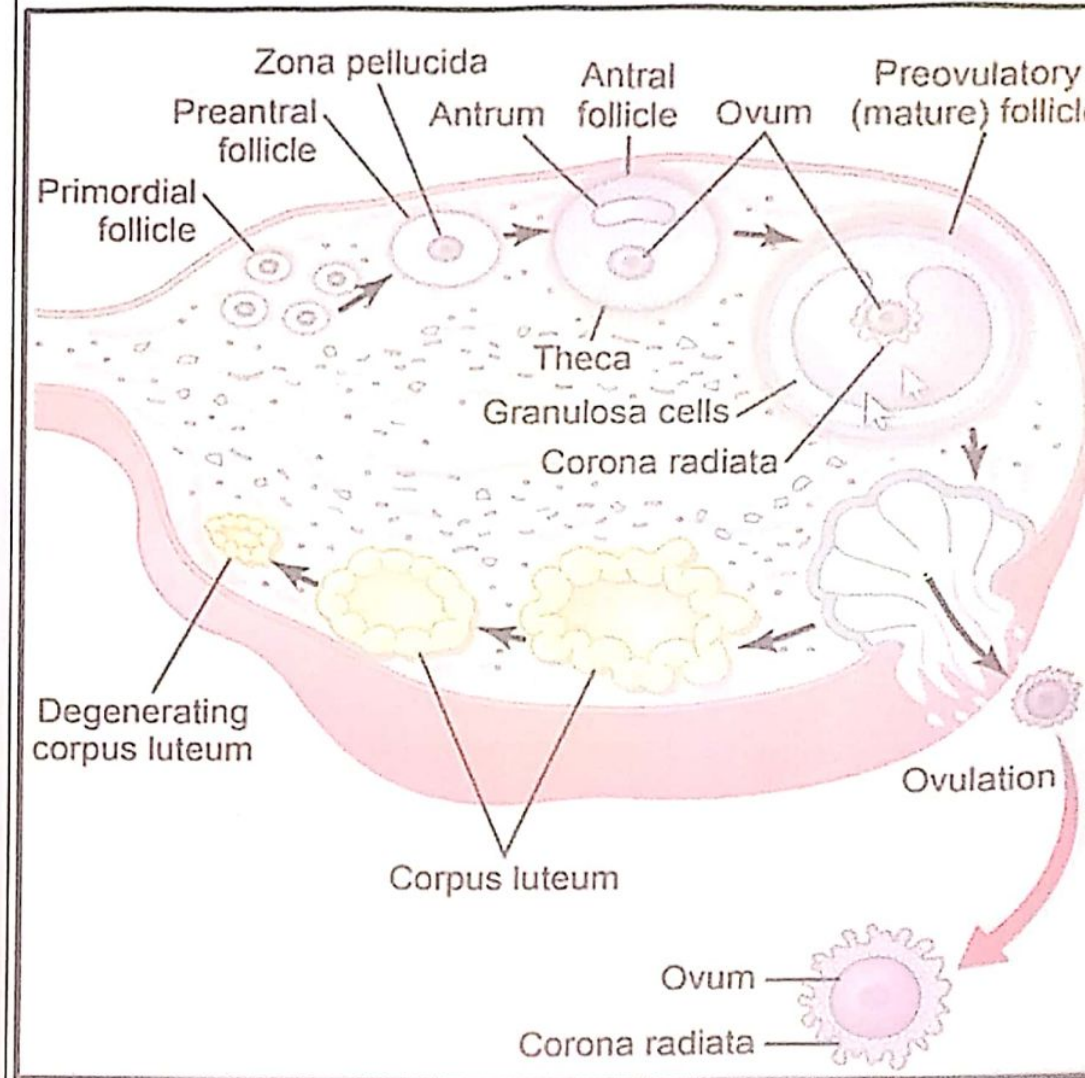


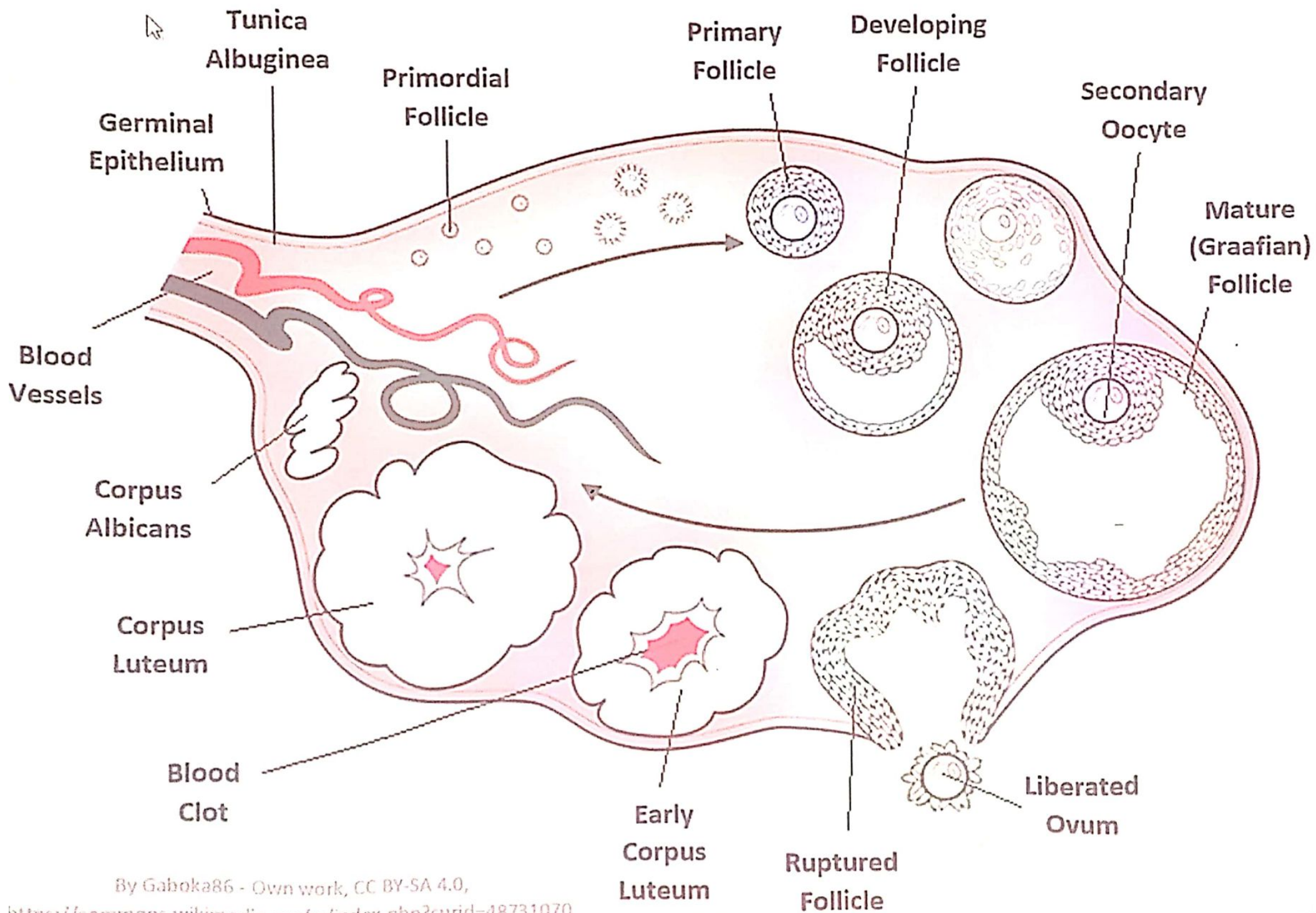
Ovulation

Due to
Large size of **mature Graafian follicle**
and **highly pressure on the ovarian wall**

↓
Ovarian wall become thin

↓
Rupture of the follicular wall
and ovarian wall.





By Gaboka86 - Own work, CC BY-SA 4.0,
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Fertilization

- union of sperm nucleus with egg nucleus to form the primary nucleus of an embryo.
- Chemical process of fertilization
 - There are chemical substances help in fertilization process that named (Gamons)
 - The sperm produce Androgamone
 - The ova produce Gynagamone

• Secretion of the sperm:

○ Androgamone I:

- ✓ Sperm cells immediately after liberation, release **Androgamone I**, the production of it continues till the arrival of sperm to the attraction zone of the egg.
- ✓ Responsible for effect of **anti-fertilizine**, which destroy the agglutination properties of fertilizine of **Gynagamone II**.
- ✓ It is responsible for retarding the excess activity of the sperm and thereby results into the conservation of energy.
- ✓ which concentrate and prolong motility of spermatozoa, it is antagonistic toward **Androgamone I** and thereby increases sperm activity

○ Androgamone II:

- ✓ Responsible for effect of **hyalorhonidase enzymes** which dissolve cement substance between cells of covering layer.

Secretion of the ova.

- Gynagamone I: Play important role for sperm activation to reach ova.
- Gynagamone II; responsible for effect of fertilizine glycoprotein specific & cause agglutination of specific sperm

Processes

- Fertilization happened in cranial third of oviduct

1. Penetrations of corona radiata: sperm penetrate it by help of

- hyaluronic hyalorhondase: secreted by Acrosomal cap of sperm to dissolve cement (hyaluronic acid) bet. corona radiata
- Whipping motion of sperm tail

2. Binding to zona pellucida

- Sperm bind to specific receptor on zona pellucida
- This binding is species-specific

3. Penetrations of zona pellucida

- Sperm form a narrow canal through zona pellucida
- Central part of Acrosomal cap form Acrosomal filament which pass through narrow canal to touch ovum cytoplasm, then form fertilization cone
- This penetration under the effect of:
 - ❖ Acrosine enzyme: to digest zona pellucida
 - ❖ Whipping motion of sperm tail

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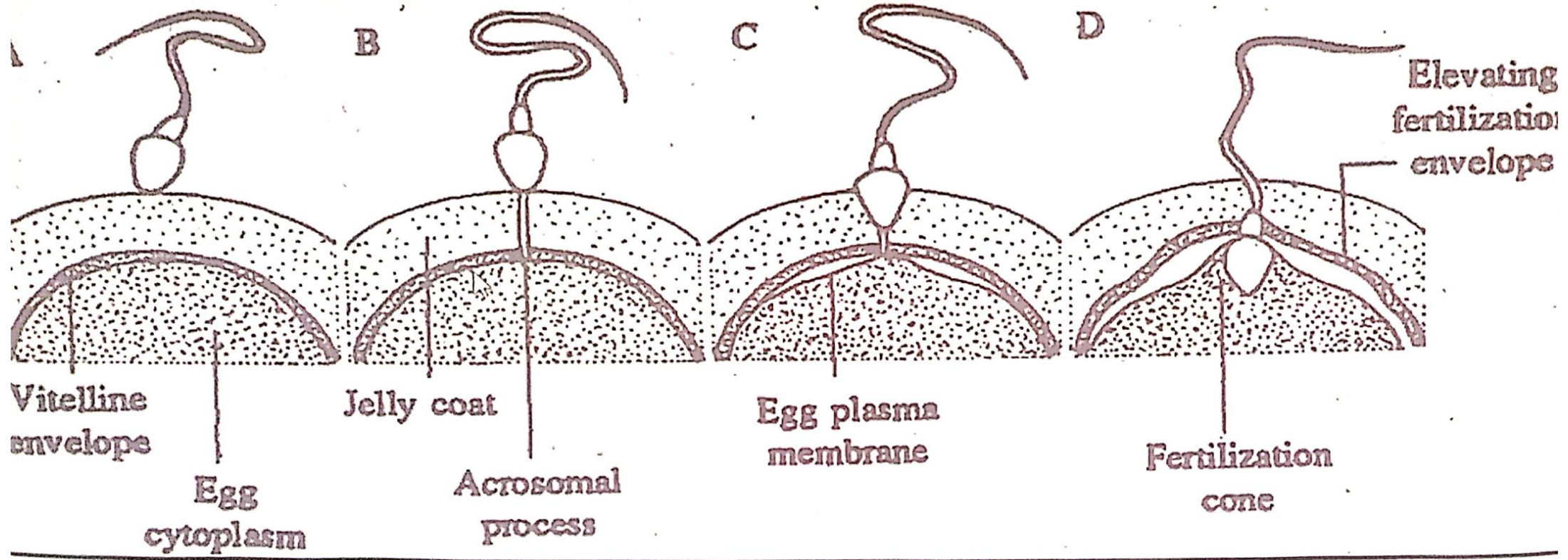
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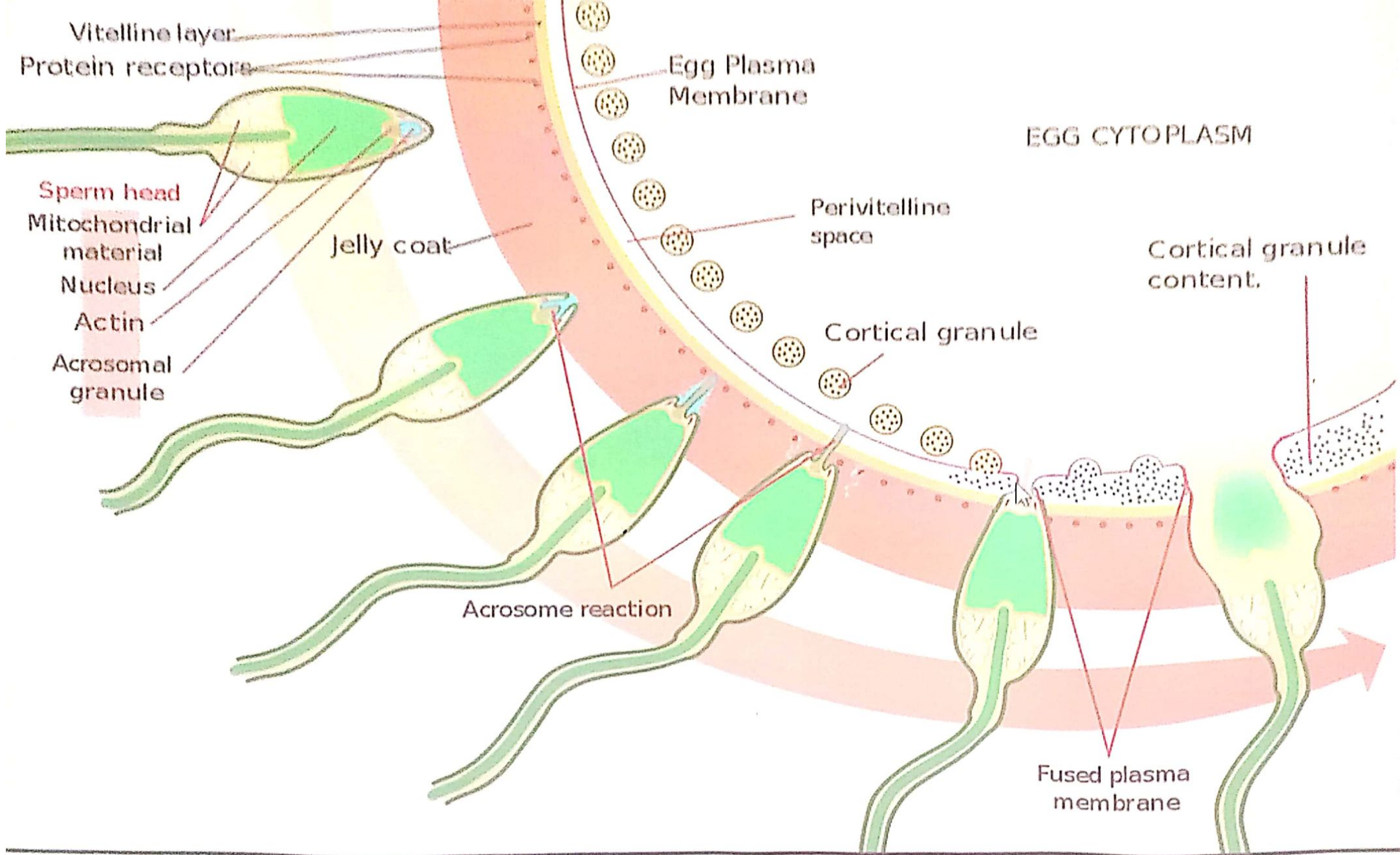
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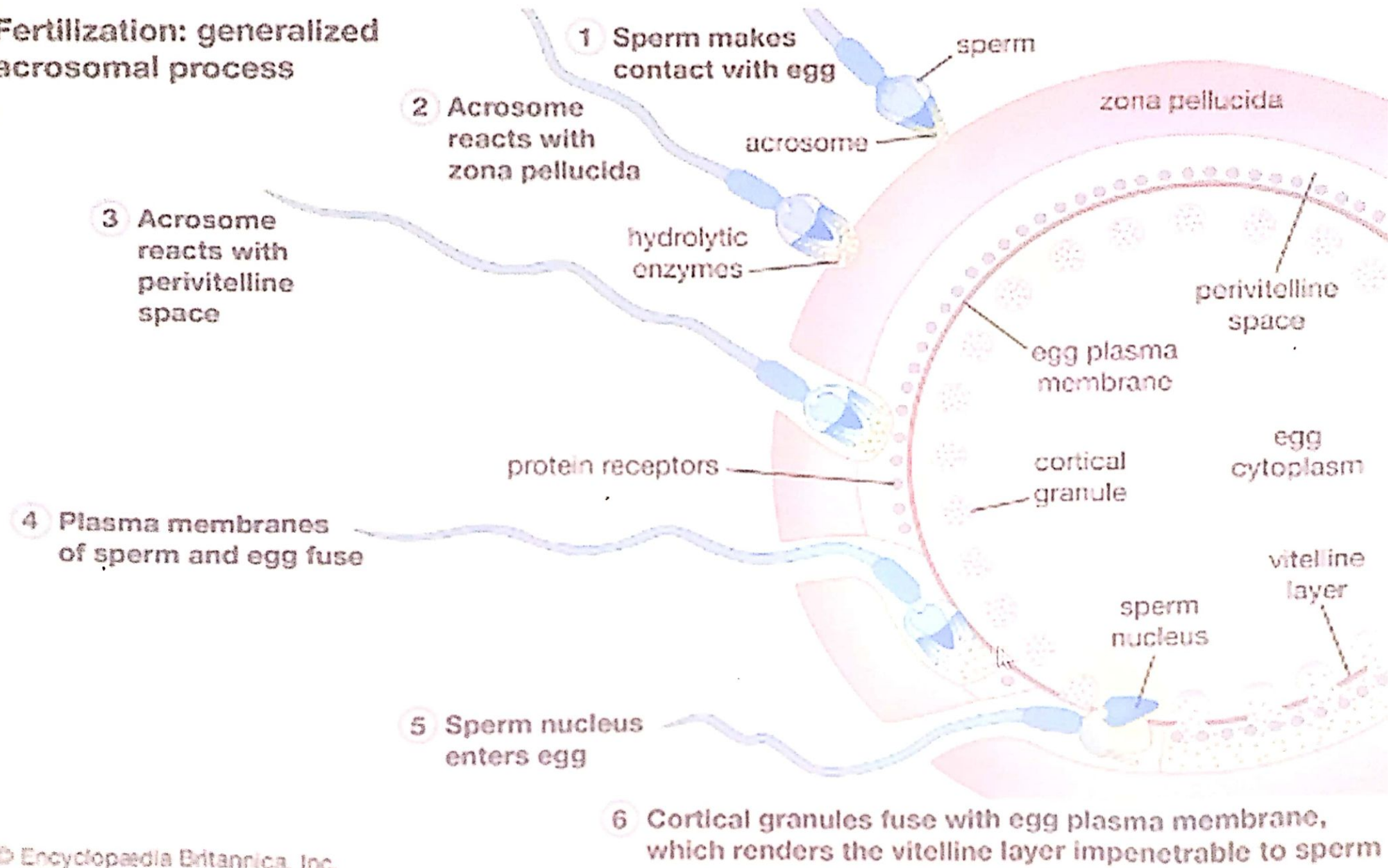
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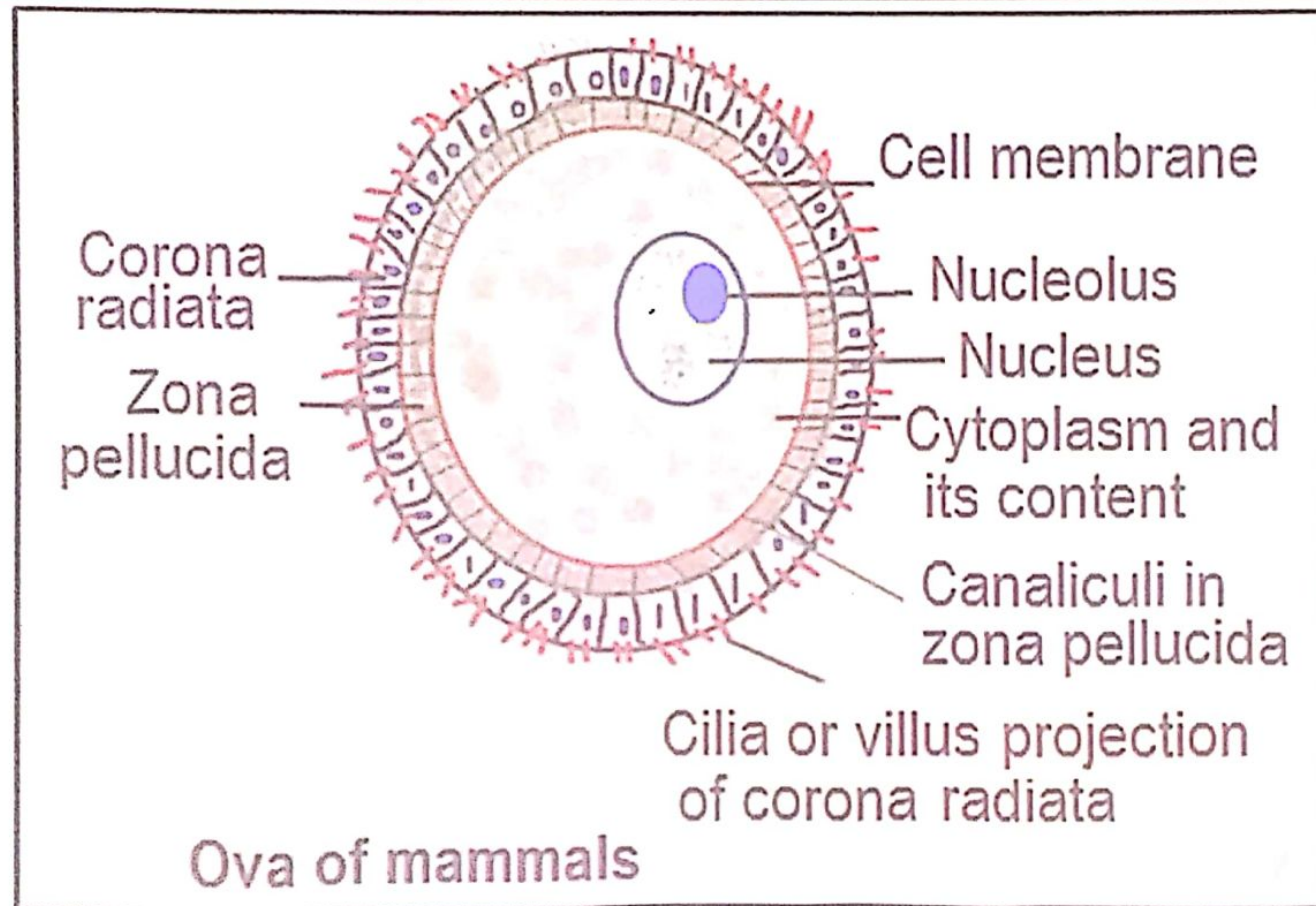
Fertilization: generalized acrosomal process



Type of ova

Ova classified according to

1. Amount of yolk.
2. Distribution of yolk
3. *According to kind of animal*



Oligo-lecithal Iso-lecithal

Amount of yolk

1- Little amount of yolk

- ✓ Oligo-lecithal or Mino-lecithal or Micro-lecithal
- ✓ Amphioxus and mammals

2- Moderate amount of yolk.

- ✓ Meso-lecithal or medio-lecithal
- ✓ Amphibia and Fishes.

3- Large amount of yolk or Huge amount of yolk.

- ✓ Poly-lecithal or Macro-lecithal or Mega-lecithal
- ✓ birds and reptile

Distribution of yolk

1-Iso-lecithal;

- ✓ Yolk uniformly distributed in the cytoplasm
- ✓ Amphioxus and Mammals

2-Telo-lecithal:

- ✓ Yolk present in vegetal pole and the cytoplasm in animal pole
- ✓ Amphibian, reptile fish and birds.

According to kind of animal

1-Oligo-lecithal Iso-lecithal :

- ✓ Small amount of yolk and equally distributed
- ✓ Mammals and Amphioxus

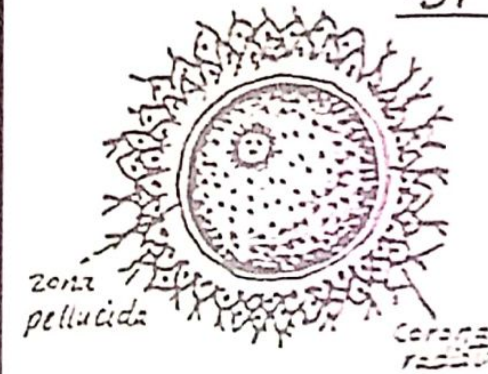
2-Medio-lecithal & Telo-lecithal :

- ✓ Moderated amount of yolk and unequal distributed
- ✓ Fishes and Amphibia

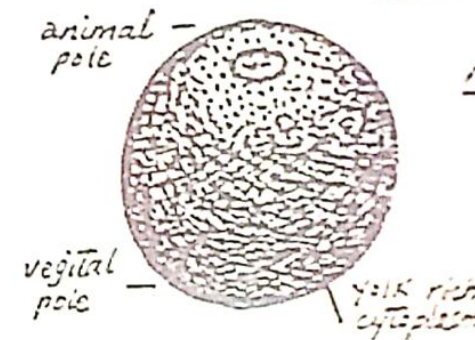
3-Poly-lecithal Telo-lecithal:

- ✓ Much amount of yolk and not equally distributed
- ✓ Reptile and birds

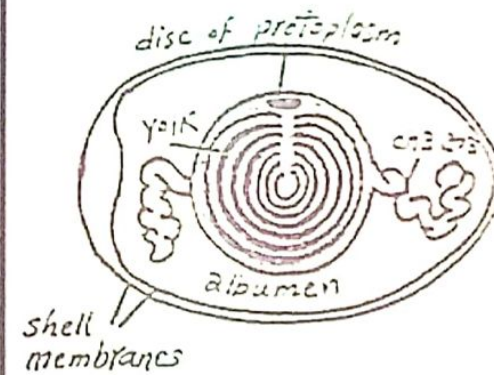
Types of Ova



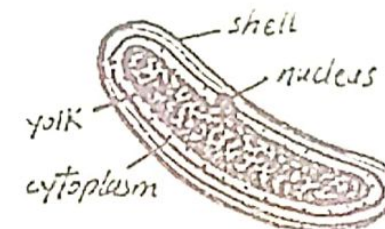
Mammalian ova
microlecithal
isolecithal



Amphibian's ova
medilecithal
telolectihal

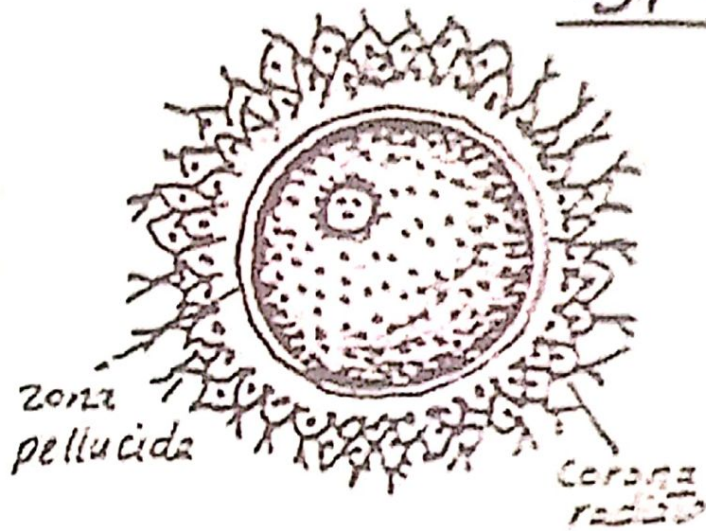


Bird's egg
macrolecithal
telolectihal



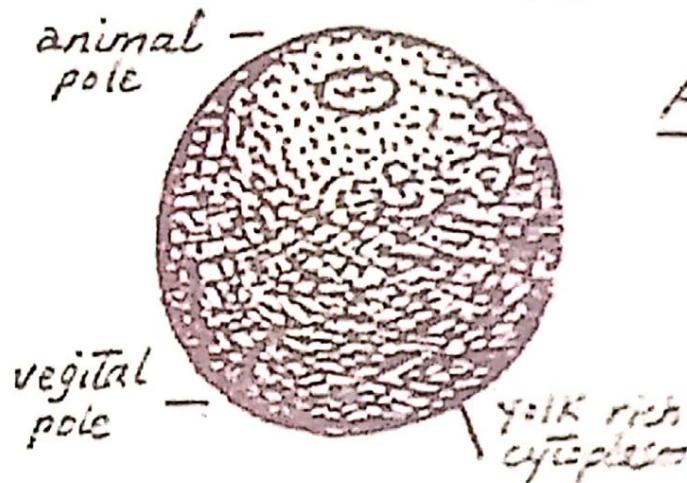
Arthropod's ova
centrolecithal

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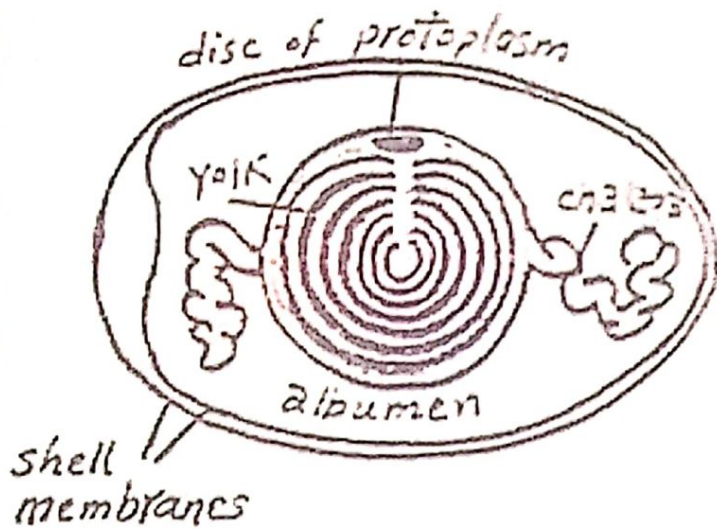
Mammalian ova

microlecithal
isolecithal



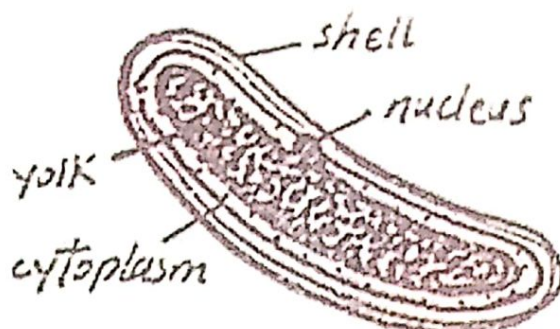
Amphibian's ova

medialcithal
telolecithal



Bird's egg

macrolecithal
telolecithal



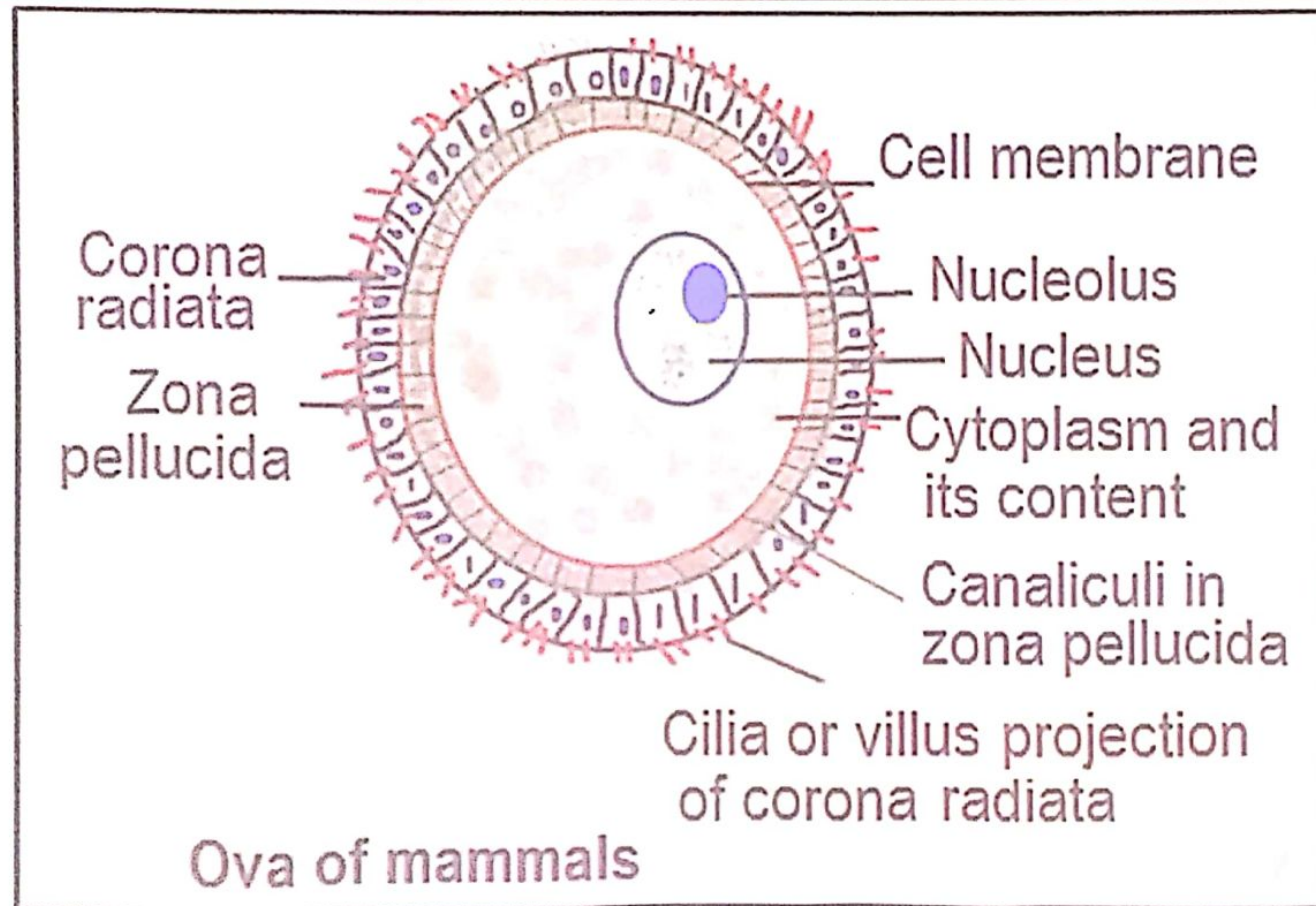
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